The Supplementary Proof Artifact

S.1. Results on Perfect KC

We explore the concept of perfect KC, which requires that the ciphertext C reveals no information about the key K. This concept is analogous to the perfect security defined by Shannon for message confidentiality [12].

Theorem S.1. *A symmetric encryption scheme* $(\mathcal{M}, \mathcal{K}, \mathcal{C}, \mathcal{E}, \mathcal{D})$ *achieves perfect KC,* if and only if *for each* $k \in \mathcal{K}$ *and* $c \in \mathcal{C}$, $p_{\mathrm{c|k}}(c \,|\, k) = p_{\mathrm{c}}(c)$, *i.e.,* $p_{\mathrm{c|k}}(c \,|\, k)$ *is independent of* k.

Proof. Echoing Shannon's proof for perfect security [12], $I(K; C) = 0$ if and only if K and C are independent, i.e., for each $k \in \mathcal{K}$ and $c \in \mathcal{C}$, $p_{\mathrm{c|k}}(c \,|\, k) = p_{\mathrm{c}}(c)$. □

This means that, for each $k \in \mathcal{K}$, E_k transforms the message distribution p_{m} into the same ciphertext distribution p_{c}. Accordingly, for each $k \in \mathcal{K}$, D_k transforms the ciphertext distribution p_{c} back into the message distribution p_{m}. Thus, the honey-message distribution under each key coincides with the real distribution. Thus, we have the following proposition.

Definition S.1. *A symmetric encryption scheme is* perfect M-IND*, if for each* $k \in \mathcal{K}$, $M_1 \equiv M_{0,k}$*, where* $C_0 \leftarrow_{p_{\mathrm{c}}} \mathcal{C}, M_{0,k} \leftarrow D_k(C_0)$ *and* $M_1 \leftarrow_{p_{\mathrm{m}}} \mathcal{M}$.

Proposition S.1. *If a symmetric encryption scheme achieves perfect KC, then it is an HE scheme with the property of perfect M-IND.*

However, unlike Theorem S.1, the converse of Proposition S.1 does not hold, because E_k can be randomized whereas D_k is deterministic. A randomized E_k may map p_{m} to different ciphertext distributions $p_{\mathrm{c|k}}(\cdot \mid k)$ under different keys, even though the corresponding D_k still maps the overall ciphertext distribution p_{c} back to p_{m}. Fig. S.2 provides such a counterexample.

Fortunately, if E_k is uniform, meaning it assigns equal probability to all possible ciphertexts in $\mathcal{C}(m, k)$ for each message m and key k, the converse of Proposition S.1 holds.

Theorem S.2. *A* uniform *symmetric encryption scheme achieves perfect KC, if and only if it is perfect M-IND.*

More interestingly, a stronger result holds: $p_{\mathrm{c|k}}(c \,|\, k)$ and $p_{\mathrm{c}}(c)$ are not only equal but also uniformly distributed over certain subspaces of $\mathcal{C}$. These subspaces are defined by a partition $\mathcal{P}$ of $\mathcal{C}$ induced by an equivalence relation R, thus $\mathcal{P} = \mathcal{C}/R$. Here, R is the equivalence closure where $c_1, c_2 \in \mathcal{C}$ are related if and only if there exist $m \in \mathcal{M}$ and $k \in \mathcal{K}$ such that $c_1, c_2 \in \mathcal{C}(m, k)$.

Proposition S.2. *If a uniform symmetric encryption scheme is perfect M-IND, then for every equivalence class* S *of* R*, the distributions* p_{c} *and* $p_{\mathrm{c|k}}$ *are identical and uniform over* S*, i.e.,* $p_{\mathrm{c}}(c_1) = p_{\mathrm{c}}(c_2) = p_{\mathrm{c|k}}(c_1 \mid k) = p_{\mathrm{c|k}}(c_2 \mid k)$ *for all* $c_1, c_2 \in S$ *and* $k \in \mathcal{K}$.

Proof of Proposition S.2. Since

$$\sum_m \Pr[D_k(C) = m] = \sum_m p_{\mathrm{m}}(m) = 1,$$

it follows that for each k, $D_k(C)$ never fails ($D_k(C) \neq \perp$).

The condition is equivalent to the following identity: for all $m \in \mathcal{M}, k \in \mathcal{K}$,

$$\begin{aligned}\sum_{c \in \mathcal{C}(m,k)} p_{\mathrm{c}}(c) = p_{\mathrm{m}}(m) \\ &= \sum_{c \in \mathcal{C}(m,k)} p_{\mathrm{m}}(m) p_{\mathrm{c|m,k}}(c \,|\, m, k) \\ &= \sum_{c \in \mathcal{C}(m,k)} p_{\mathrm{c|k}}(c \,|\, k).\end{aligned}$$

Here, $p_{\mathrm{c|k}}(c \,|\, k)$ is constant over $\mathcal{C}(m, k)$, given by

$$p_{\mathrm{c|k}}(c \,|\, k) = \frac{p_{\mathrm{m}}(m)}{|\mathcal{C}(m, k)|}.$$

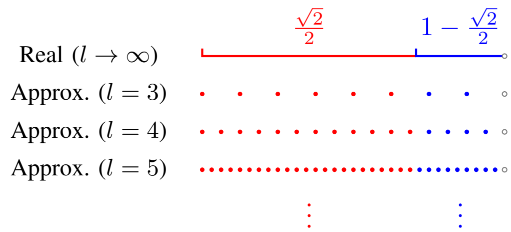

Figure S.1: Approximating the probability distribution $\left(\frac{\sqrt{2}}{2}, 1 - \frac{\sqrt{2}}{2}\right)$ over $\{m_1, m_2\}$ by random l bits (i.e., l-bit ciphertexts/ seeds).

Intuitively, $p_{\mathrm{c}}(c)$ is the weighted average of $p_{\mathrm{c}|\mathrm{k}}(c \mid k)$ over all k, with weights $p_{\mathrm{k}}(k)$. Typically, the maximum of $p_{\mathrm{c}|\mathrm{k}}(c \,|\, k)$ is greater than the average $p_{\mathrm{c}}(c)$. However, the equation above implies that the maximum is equal to the average, forcing $p_{\mathrm{c}|\mathrm{k}}(c \,|\, k) = p_{\mathrm{c}}(c)$. Based on this intuition, we now present a formal proof.

Formally, consider any equivalence class S of R, and let c_1, k_1 be the pair that maximizes $p_{\mathrm{c}|\mathrm{k}}(c \,|\, k)$ over all $k \in \mathcal{K}, c \in S$. Define $m_1 = D_{k_1}(c_1)$.

By the uniformity of encryption, for all $c \in \mathcal{C}(m_1, k_1)$,

$$p_{\mathrm{c}|\mathrm{k}}(c \,|\, k_1) = p_{\mathrm{c}|\mathrm{k}}(c_1 \,|\, k_1) = \frac{p_{\mathrm{m}}(m_1)}{|\mathcal{C}(m_1, k_1)|}.$$

Therefore, we derive the following:

$$\begin{aligned}
\sum_{c \in \mathcal{C}(m_1, k_1)} p_{\mathrm{c}|\mathrm{k}}(c \,|\, k_1) &= \sum_{c \in \mathcal{C}(m_1, k_1)} p_{\mathrm{c}}(c) \\
&= \sum_{c \in \mathcal{C}(m_1, k_1)} \sum_{k} p_{\mathrm{k}}(k) p_{\mathrm{c}|\mathrm{k}}(c \,|\, k) \\
&\leq \sum_{c \in \mathcal{C}(m_1, k_1)} \sum_{k} p_{\mathrm{k}}(k) p_{\mathrm{c}|\mathrm{k}}(c \,|\, k_1) \\
&= \sum_{c \in \mathcal{C}(m_1, k_1)} p_{\mathrm{c}|\mathrm{k}}(c \,|\, k_1).
\end{aligned}$$

Since both sides are equal, the inequality must hold with equality, forcing

$$p_{\mathrm{c}|\mathrm{k}}(c \,|\, k) = p_{\mathrm{c}|\mathrm{k}}(c \,|\, k_1) = p_{\mathrm{c}|\mathrm{k}}(c_1 \,|\, k_1)$$

for all $k \in \mathcal{K}$ and $c \in \mathcal{C}(m_1, k_1)$. Furthermore, for all $c \in \mathcal{C}(m_1, k_1)$,

$$p_{\mathrm{c}}(c) = \sum_{k} p_{\mathrm{k}}(k) p_{\mathrm{c}|\mathrm{k}}(c \,|\, k) = p_{\mathrm{c}|\mathrm{k}}(c_1 \,|\, k_1).$$

This result shows that the maximum value $p_{\mathrm{c}|\mathrm{k}}(c_1 \mid k_1)$ propagates to all $p_{\mathrm{c}}(c)$ and $p_{\mathrm{c}|\mathrm{k}}(c \mid k)$ within $\mathcal{C}(m_1, k_1)$. By the definition of equivalence classes, this propagation extends to the entire S through the same argument. In other words, p_{c} and $p_{\mathrm{c}|\mathrm{k}}$ are identical and uniform over S. $\square$

Impossibility of perfect KC. For a general message distribution, such as the two-message distribution $\left(\frac{\sqrt{2}}{2}, 1 - \frac{\sqrt{2}}{2}\right)$ illustrated in Fig. S.1, it is impossible to achieve perfect KC with a finite ciphertext length l and uniform encryption, except for trivial encryption schemes. Specifically, in these trivial schemes, E_k is the identity mapping id or any mapping independent of k, ensuring perfect KC. However, such trivial schemes fail to provide message confidentiality, as the ciphertext completely determines the message without requiring the key. Except for these trivial schemes, we establish by contradiction that no uniform encryption scheme can achieve perfect KC.

Assume that a uniform encryption scheme achieves perfect KC. Then, by Proposition S.2, $p_{\mathrm{c}} = p_{\mathrm{c}|\mathrm{k}}(\cdot \mid k)$ must be uniform over each equivalence class $S \in \mathcal{C}/R$. Thus, for any $\mathcal{C}(m, k)$ and $\mathcal{C}(m', k')$ belonging to the same S,

$$\frac{p_{\mathrm{m}}(m)}{p_{\mathrm{m}}(m')} = \frac{|\mathcal{C}(m, k)|}{|\mathcal{C}(m', k')|} \in \mathbb{Q}.$$

However, since $\frac{p_{\mathrm{m}}(m_1)}{p_{\mathrm{m}}(m_2)} = 1 + \sqrt{2} \notin \mathbb{Q}$, it follows that $\mathcal{C}(m_1, k)$ and $\mathcal{C}(m_2, k')$ cannot belong to the same S for any k, k'. Thus, $\mathcal{C}(m_1, k) \cap \mathcal{C}(m_2, k') = \emptyset$. Further, by $\mathcal{C}(m_1, k) \cup \mathcal{C}(m_2, k) = \mathcal{C}$, $\mathcal{C}(m_1, k)$ and $\mathcal{C}(m_2, k)$ remain unchanged across different keys, which are the only two equivalence classes of $\mathcal{C}$. This implies that the encryption scheme is trivial.

A non-uniform (and non-trivial) encryption scheme may achieve perfect KC, but it must assign *irrational* probabilities to ciphertexts, which still requires an infinite number of uniform random bits.

S.2. Bounds for HE schemes with Specific Well-designed Encoders

IS-DTE. Juels and Ristenpart [10] introduced a general approach to build encoders for arbitrary non-uniform distributions, resulting in what are known as IS-DTEs. For given p_{m} over $\mathcal{M}$, we denote its cumulative distribution function by F, where $F(i) = \sum_{t=1}^{i} p_{\mathrm{m}}(m_t)$. The construction of IS-DTE leverages the fact that, for a uniform random variable U on $[0, 1)$, the inverse function $F^{-1}(U)$ follows the distribution p_{m}. By discretizing $[0, 1)$ to the set $\mathcal{S} = \{0, 1, \dots, 2^l - 1\}$ (i.e., $\{0, 1\}^l$), IS-DTE randomly assigns an integer seed in the range $[\mathrm{round}(2^l F(i-1)), \mathrm{round}(2^l F(i)))$ to the message m_i. Here, round represents the rounding function, with $F(0) = 0$, and $l \geq \lceil -\log \min_m p_{\mathrm{m}}(m) \rceil$[1]. With this design, $|p_{\mathrm{d}}(m) - p_{\mathrm{m}}(m)| \leq \frac{1}{2^l}$ for each $m \in \mathcal{M}$.

Corollary S.1. *Built on a cipher and an IS-DTE with the MS-IND property, the HE scheme achieves KC. Specifically,*

$$I(K; C^{(l)}) \leq \frac{2p_{\mathrm{m}}^{\min} - \frac{1}{2^l}}{2^l \ln 2 p_{\mathrm{m}}^{\min} \left(p_{\mathrm{m}}^{\min} - \frac{1}{2^l}\right)} = O\left(\frac{1}{2^l}\right).$$

For messages with complex distributions, such as natural language texts, directly applying IS-DTE often leads to prohibitive time and space complexity. To address this, Cheng et al. [28] proposed a method to construct an efficient encoder based on a probability model, termed the probability-model-transforming encoder (PMTE). Similar to IS-DTE, IS-PMTE is designed using inverse sampling. During encoding, IS-PMTE parses the message into a sequence of generating rules and encodes each rule individually. If the probability model accurately characterizes the message distribution, IS-PMTE achieves MS-IND. Without accounting for the differences between the probability model and the actual message distribution, $|p_{\mathrm{d}}(m) - p_{\mathrm{m}}(m)| \leq \frac{l_{\mathrm{seq}} n_{\mathrm{r}}}{2^{l/l_{\mathrm{seq}}}}$ holds for each message m, where l_{seq} represents the maximum sequence length and n_{r} denotes the number of rules.

Corollary S.2. *Built on a cipher and an IS-PMTE with the MS-IND property, the HE scheme achieves KC. Specifically,*

$$I(K; C^{(l)}) \leq \frac{l_{\mathrm{seq}} n_{\mathrm{r}} \left(2p_{\mathrm{m}}^{\min} - \frac{l_{\mathrm{seq}} n_{\mathrm{r}}}{2^{l/l_{\mathrm{seq}}}}\right)}{2^{l/l_{\mathrm{seq}}} \ln 2\, p_{\mathrm{m}}^{\min} \left(p_{\mathrm{m}}^{\min} - \frac{l_{\mathrm{seq}} n_{\mathrm{r}}}{2^{l/l_{\mathrm{seq}}}}\right)} = O\left(2^{-l/l_{\mathrm{seq}}}\right).$$

S.3. Proofs in Section 4

Theorem S.3 (Pinsker's inequality [26]). *For two random variables X_0 and X_1 on the space $\mathcal{X}$ with the distributions p_0 and p_1,*

$$\mathrm{SD}(X_0, X_1) \;\leq\; \sqrt{\frac{\ln 2}{2} \mathrm{KL}(X_0 \| X_1)}\,.$$

Theorem S.4 (Theorem 7 in [27]). *For two random variables X_0 and X_1 on the space $\mathcal{X}$ with the distributions p_0 and p_1 such that for each $x \in \mathcal{X}$, $p_0(x) \leq c \cdot p_1(x)$, where $c \geq 1$ is a constant:*

$$\mathrm{KL}(X_0 || X_1) \leq \frac{c \log c}{c - 1} \mathrm{SD}(X_0, X_1).$$

Theorem 1 (restated). *A symmetric encryption scheme achieves statistical KC if and only if it satisfies C-IND in Fig. 3, i.e., for every key k we have $C_1^{(l)} \overset{s}{\equiv} C_{0,k}^{(l)}$. More specifically,*

$$\mathrm{SD}(C_{0,k}^{(l)}, C_1^{(l)}) \leq \sqrt{\frac{\ln 2}{2p_{\mathrm{k}}^{\min}} I(K; C^{(l)})} = O\left(\sqrt{I(K; C^{(l)})}\right), \ \forall k \in \mathcal{K};$$

$$I(K; C^{(l)}) \leq \frac{\log(1/p_{\mathrm{k}}^{\min})}{1 - p_{\mathrm{k}}^{\min}} \max_k \mathrm{SD}(C_{0,k}^{(l)}, C_1^{(l)}) = O\left(\max_k \mathrm{SD}(C_{0,k}^{(l)}, C_1^{(l)})\right).$$

1. $\lceil x \rceil$ represents the smallest integer greater than or equal to x.

Proof of Theorem 1. Note $I(K;C^{(l)}) = \mathbb{E}_{K\leftarrow_{p_\mathrm{k}}\mathcal{K}}[\mathrm{KL}(C_{0,K}^{(l)}||C_1^{(l)})]$, since

$$\begin{aligned}
I(K;C^{(l)}) &= \sum_{k\in\mathcal{K},c\in\mathcal{C}^{(l)}} p_\mathrm{k}(k)p_{\mathrm{c|k}}(c\,|\,k)\log\frac{p_\mathrm{k}(k)p_{\mathrm{c|k}}(c\,|\,k)}{p_\mathrm{k}(k)p_\mathrm{c}(c)}\\
&= \sum_{k\in\mathcal{K}} p_\mathrm{k}(k)\sum_{c\in\mathcal{C}^{(l)}} p_{\mathrm{c|k}}(c)\log\frac{p_{\mathrm{c|k}}(c\,|\,k)}{p_\mathrm{c}(c)}\\
&= \sum_{k\in\mathcal{K}} p_\mathrm{k}(k)\mathrm{KL}(C_{0,k}^{(l)}||C_1^{(l)})\\
&= \mathop{\mathbb{E}}_{K\leftarrow_{p_\mathrm{k}}\mathcal{K}}[\mathrm{KL}(C_{0,K}^{(l)}||C_1^{(l)})].
\end{aligned}$$

With the equation, we can easily prove the necessity (i.e., the negligibility of $I(K;C^{(l)})$ implies that of $\mathrm{SD}(C_{0,k}^{(l)},C_1^{(l)})$ for each $k\in\mathcal{K}$):

Since $I(K;C^{(l)}) = \mathbb{E}_{K\leftarrow_{p_\mathrm{k}}\mathcal{K}}[\mathrm{KL}(C_{0,k}^{(l)}||C_1^{(l)})]$, for each $k\in\mathcal{K}$, $\mathrm{KL}(C_{0,k}^{(l)}||C_1^{(l)}) \leq \frac{I(K;C^{(l)})}{p_\mathrm{k}^{\min}}$. Further, by Pinsker's inequality,

$$\begin{aligned}
\mathrm{SD}(C_{0,k}^{(l)},C_1^{(l)}) &\leq \sqrt{\frac{\ln 2}{2}\mathrm{KL}(C_{0,k}^{(l)}||C_1^{(l)})}\\
&\leq \sqrt{\frac{\ln 2}{2p_\mathrm{k}^{\min}}\cdot I(K;C^{(l)})}.
\end{aligned}$$

Thus, the necessity holds.

By Theorem S.4, we naturally get the sufficiency (i.e., the negligibility of $\mathrm{SD}(C_{0,k}^{(l)},C_1^{(l)})$ for each $k\in\mathcal{K}$ implies that of $I(K;C^{(l)})$):

Note that for each $k\in\mathcal{K}$, $p_\mathrm{c}(c)=\sum_{k\in\mathcal{K}}p_\mathrm{k}(k)p_{\mathrm{c|k}}(c\,|\,k)$, and thus

$$p_{\mathrm{c|k}}(c\,|\,k)\leq p_\mathrm{c}(c)\cdot\frac{1}{p_\mathrm{k}^{\min}},$$

where $\frac{1}{p_\mathrm{k}^{\min}}$ is a constant. Applying a reverse Pinsker's inequality, i.e., Theorem S.4 (X_0 and X_1 correspond to $C_{0,k}^{(l)}$ and $C_1^{(l)}$, meanwhile p_0 and p_1 correspond to $p_{\mathrm{c|k}}$ and p_c),

$$\mathrm{KL}(C_{0,k}^{(l)}||C_1^{(l)})\leq\frac{\log\frac{1}{p_\mathrm{k}^{\min}}}{1-p_\mathrm{k}^{\min}}\mathrm{SD}(C_{0,k}^{(l)},C_1^{(l)})$$

Further,

$$I(K;C^{(l)})\leq\frac{\log\frac{1}{p_\mathrm{k}^{\min}}}{1-p_\mathrm{k}^{\min}}\max_k\mathrm{SD}(C_{0,k}^{(l)},C_1^{(l)})$$

Thus, the sufficiency holds. □

Theorem 2 (restated). *A* uniform *symmetric encryption scheme achieves statistical KC, if and only if it is M-IND. Specifically,*

$$\max_k\mathrm{SD}(C_{0,k},C_1)\leq\left(2+\frac{1}{p_\mathrm{k}^{\min}}\right)^{|\mathcal{M}|\cdot|\mathcal{K}|}\max_k\mathrm{SD}(M_{0,k},M_1).$$

Proof of Theorem 2. We prove sufficiency; necessity follows from Proposition 1.

By Proposition 1, it suffices to bound $\mathrm{SD}(C_{0,k},C_1)=\sum_{c\in\mathcal{C}}\left|p_\mathrm{c}(c)-p_{\mathrm{c|k}}(c\,|\,k)\right|$ for all $k\in\mathcal{K}$.

We partition the ciphertext space $\mathcal{C}$ into parts, i.e., $\mathcal{C}=\cup_i\mathcal{C}_i$. We exclude ciphertexts c with $p_\mathrm{c}(c)=0$ from $\mathcal{C}$, as they do not contribute to the statistical distance. For each $\mathcal{C}_i$, we aim to bound

$$\rho_{i,k}=\sum_{c\in\mathcal{C}_i}\left|p_\mathrm{c}(c)-p_{\mathrm{c|k}}(c\,|\,k)\right|.$$

Note that $D_k(C^{(l)})$ may fail and output $\perp$, thus the following analysis is on the extended message space $\mathcal{M}'=\mathcal{M}\cup\{\perp\}$ with $p_\mathrm{m}(\perp)=0$.

Let c_1,k_1 be the pair that maximizes $p_{\mathrm{c|k}}(c\,|\,k)$ over all $k\in\mathcal{K},c\in\mathcal{C}$, and let $m_1=D_{k_1}(c_1)$. Define $\mathcal{C}_1=\mathcal{C}(m_1,k_1)$.

Iteratively, for each $i \geq 2$, let c_i, k_i be the pair that maximizes $p_{\mathrm{c}|\mathrm{k}}(c\,|\,k)$ over all $k \in \mathcal{K}$ and $c \in \mathcal{C} - \bigcup_{j=1}^{i-1} \mathcal{C}_j$, and let $m_i = D_{k_i}(c_i)$. Define $\mathcal{C}_i = \mathcal{C}(m_i, k_i) - \bigcup_{j=1}^{i-1} \mathcal{C}_j$.

Repeat this construction until $\bigcup_{i=1}^{t} \mathcal{C}_i = \mathcal{C}$, where t is the number of iterations. Since the family $\{\mathcal{C}(m,k)\}_{m\in\mathcal{M},\, k\in\mathcal{K}}$ covers $\mathcal{C}$ (we exclude ciphertexts c with $p_{\mathrm{c}}(c) = 0$), we can select at most $|\mathcal{M}|\cdot|\mathcal{K}|$ distinct pairs (m_i, k_i). Hence $t \leq |\mathcal{M}|\cdot|\mathcal{K}|$.

Next, we bound $\rho_{i,k}$ separately. For each $i = 1, 2, \ldots, t$ and for all $c \in \mathcal{C}_i$, $k \in \mathcal{K}$,

$$p_{\mathrm{c}|\mathrm{k}}(c\,|\,k) \leq p_{\mathrm{c}|\mathrm{k}}(c_i\,|\,k_i) = p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i).$$

Therefore $p_{\mathrm{c}|\mathrm{k}}(c\,|\,k) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i) \leq 0$. For any other $k' \in \mathcal{K} - \{k_i\}$, we have:

$$\begin{aligned}
\rho_{i,k_i} = \sum_{c\in\mathcal{C}_i} \left|p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\right| &= \sum_{c\in\mathcal{C}_i} \left|\sum_{k} p_{\mathrm{k}}(k) p_{\mathrm{c}|\mathrm{k}}(c\,|\,k) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\right| \\
&= \sum_{c\in\mathcal{C}_i} \left|\sum_{k} p_{\mathrm{k}}(k) \Big(p_{\mathrm{c}|\mathrm{k}}(c\,|\,k) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\Big)\right| \\
&= \sum_{k} p_{\mathrm{k}}(k) \sum_{c\in\mathcal{C}_i} \left|p_{\mathrm{c}|\mathrm{k}}(c\,|\,k) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\right| \\
&\geq p_{\mathrm{k}}(k') \sum_{c\in\mathcal{C}_i} \left|p_{\mathrm{c}|\mathrm{k}}(c\,|\,k') - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\right|.
\end{aligned}$$

Thus:

$$\begin{aligned}
\rho_{i,k'} &= \sum_{c\in\mathcal{C}_i} \left|p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k')\right| \\
&\leq \sum_{c\in\mathcal{C}_i} \left|p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\right| \\
&\quad + \sum_{c\in\mathcal{C}_i} \left|p_{\mathrm{c}|\mathrm{k}}(c\,|\,k') - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\right| \\
&\leq \left(1 + \frac{1}{p_{\mathrm{k}}(k')}\right) \sum_{c\in\mathcal{C}_i} \left|p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\right| \\
&= \left(1 + \frac{1}{p_{\mathrm{k}}(k')}\right) \rho_{i,k_i}.
\end{aligned}$$

Thus $\rho_{i,k'}$ is bounded by ρ_{i,k_i}, so it only suffices to bound ρ_{i,k_i}. Again, by maximality of $p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)$,

$$p_{\mathrm{c}}(c) = \sum_{k} p_{\mathrm{k}}(k) p_{\mathrm{c}|\mathrm{k}}(c\,|\,k) \leq p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i).$$

so $p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)$ also remains non-positive. Let $d = \max_k \mathrm{SD}(M_{0,k}, M_1)$. For ρ_{1,k_1}, we obtain:

$$\begin{aligned}
\rho_{1,k_1} &= \sum_{c\in\mathcal{C}(m_1,k_1)} \left|p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_1)\right| \\
&= \left|\sum_{c\in\mathcal{C}(m_1,k_1)} \Big(p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_1)\Big)\right| \\
&= \Big|\Pr[D_{k_1}(C_1) = m_1] - p_{\mathrm{m}}(m_1)\Big| \leq d.
\end{aligned}$$

For ρ_{i,k_i} with $2 \leq i \leq t$,

$$\begin{aligned}
\rho_{i,k_i} &= \sum_{c\in\mathcal{C}_i} \left|p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\right| \\
&= \left|\sum_{c\in\mathcal{C}_i} \Big(p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\Big)\right| \\
&= \left|\sum_{c\in\mathcal{C}(m_i,k_i)} \Big(p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c\,|\,k_i)\Big)\right.
\end{aligned}$$

$$
\begin{aligned}
&-\sum_{j=1}^{i-1} \sum_{c \in \mathcal{C}(m_i,k_i)\cap \mathcal{C}_j} \Big(p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c|k_i)\Big)\Bigg| \\
\le & \left| \sum_{c \in \mathcal{C}(m_i,k_i)} \Big(p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c|k_i)\Big)\right| \\
&+\sum_{j=1}^{i-1} \left| \sum_{c \in \mathcal{C}(m_i,k_i)\cap \mathcal{C}_j} \Big(p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c|k_i)\Big)\right| \\
\le & \Big| \Pr[D_{k_i}(C_1) = m_i] - p_{\mathrm{m}}(m_i) \Big| + \sum_{j=1}^{i-1} \sum_{c \in \mathcal{C}_j} \big| p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c|k_i) \big| \\
\le & \, d + \sum_{j=1}^{i-1} \rho_{j,k_i} \le d + \left(1 + \frac{1}{p_{\mathrm{k}}(k_i)}\right) \sum_{j=1}^{i-1} \rho_{j,k_j}.
\end{aligned}
$$

Define $\mu_i = \sum_{j=1}^{i} \rho_{j,k_j}$, we have $\mu_1 = \rho_{1,k_1} \le d$ and $\mu_i - \mu_{i-1} \le d + \left(1 + \frac{1}{p_{\mathrm{k}}(k_i)}\right) \mu_{i-1}$. Then,

$$
\mu_i \le d + \left(2 + \frac{1}{p_{\mathrm{k}}(k_i)}\right) \mu_{i-1} \le d + \left(2 + \frac{1}{p_{\mathrm{k}}^{\min}}\right) \mu_{i-1}.
$$

Unrolling the recurrence, we obtain

$$
\mu_i \le \frac{p_{\mathrm{k}}^{\min}}{1 + p_{\mathrm{k}}^{\min}} \left(\left(2 + \frac{1}{p_{\mathrm{k}}^{\min}}\right)^i - 1 \right) d.
$$

Finally, for all $k \in \mathcal{K}$, we can bound $\mathrm{SD}(C_{0,k}, C_1)$:

$$
\begin{aligned}
\mathrm{SD}(C_{0,k}, C_1) &= \sum_{c \in \mathcal{C}} \big| p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c|k) \big| \\
&= \sum_{i=1}^{t} \sum_{c \in \mathcal{C}_i} \big| p_{\mathrm{c}}(c) - p_{\mathrm{c}|\mathrm{k}}(c|k) \big| \\
&= \sum_{i=1}^{t} \rho_{i,k} \le \sum_{i=1}^{t} \left(1 + \frac{1}{p_{\mathrm{k}}(k)}\right) \rho_{i,k_i} \\
&\le \left(1 + \frac{1}{p_{\mathrm{k}}^{\min}}\right) \mu_t \le \left(\left(2 + \frac{1}{p_{\mathrm{k}}^{\min}}\right)^t - 1 \right) d \\
&\le \left(2 + \frac{1}{p_{\mathrm{k}}^{\min}}\right)^{|\mathcal{M}|\cdot|\mathcal{K}|} \max_k \mathrm{SD}(M_{0,k}, M_1),
\end{aligned}
$$

which finishes the proof. □

Proposition 2 (restated). *If a uniform symmetric encryption scheme is M-IND, then for each $k, k' \in \mathcal{K}$, $C_{0,k}^{(l)} \overset{s}{\equiv} C_1^{(l)} \overset{s}{\equiv} U_{k'}^{(l)}$, where $C_{0,k}^{(l)} \leftarrow_{p_{\mathrm{c}|\mathrm{k}}(\cdot|k)} \mathcal{C}^{(l)}$, $C_1^{(l)} \leftarrow_{p_{\mathrm{c}}} \mathcal{C}^{(l)}$, and $U_{k'}^{(l)}$ is uniform over each $S \in P_{k'}^{(l)}$.*

Proof of Proposition 2. It is naturally implied by Theorem 2. □

Theorem 3 (restated). *An EE construction with an encoder and a cipher achieves statistical KC if the encoder is MS-IND. Specifically, with $l \ge \log 1/p_{\mathrm{m}}^{\min} + 1$,*

$$
I(K; C^{(l)}) \le \frac{\mathrm{SD}(M_0^{(l)}, M_1)(2p_{\mathrm{m}}^{\min} - \mathrm{SD}(M_0^{(l)}, M_1))}{\ln 2 \cdot p_{\mathrm{m}}^{\min}(p_{\mathrm{m}}^{\min} - \mathrm{SD}(M_0^{(l)}, M_1))} = O(\mathrm{SD}(M_0^{(l)}, M_1)).
$$

where $M_1 \leftarrow_{p_{\mathrm{m}}} \mathcal{M}$ and $M_0^{(l)} \leftarrow_{p_{\mathrm{d}}} \mathcal{M}$.

Proof of Theorem 3. With the approach in the proof of Theorem 1, we aim to bound $I(K; C^{(l)})$ by $\mathrm{KL}(C_{0,k}^{(l)} || C_1^{(l)})$, as $I(K; C^{(l)}) = \mathbb{E}_{K \leftarrow_{p_{\mathrm{k}}} \mathcal{K}}[\mathrm{KL}(C_{0,K}^{(l)} || C_1^{(l)})]$, where $C_{0,k}^{(l)} \leftarrow_{p_{\mathrm{c}|\mathrm{k}}(\cdot|k)} \mathcal{C}^{(l)}$ and $C_1^{(l)} \leftarrow_{p_{\mathrm{c}}} \mathcal{C}^{(l)}$.

Further, since $\mathrm{KL}(C_{0,k}^{(l)}||C_1^{(l)}) = \sum_{c\in\mathcal{C}^{(l)}} p_{\mathrm{c|k}}(c\,|\,k)\log\frac{p_{\mathrm{c|k}}(c|k)}{p_{\mathrm{c}}(c)}$, our strategy focuses on constraining $\mathrm{KL}(C_{0,k}^{(l)}||C_1^{(l)})$ by $\frac{p_{\mathrm{c|k}}(c|k)}{p_{\mathrm{c}}(c)}$.

Denoting the encoding scheme as $(\mathtt{encode}^{(l)}, \mathtt{decode}^{(l)})$ and the cipher as $(\mathtt{encrypt}^{(l)}, \mathtt{decrypt}^{(l)})$, we have

$$\begin{aligned}
p_{\mathrm{c|k}}(c\,|\,k) &= \Pr[C^{(l)} = c\,|\,K = k] \\
&= \Pr\Big[C^{(l)} = c,\, S^{(l)} = \mathtt{decrypt}_k^{(l)}(C^{(l)}),\, M = \mathtt{decode}^{(l)}(S^{(l)})\,|\,K = k\Big] \\
&= \Pr[M = D_k^{(l)}(c)] \\
&\quad\cdot \Pr[S^{(l)} = \mathtt{decrypt}_k^{(l)}(C^{(l)})\,|\,M = D_k^{(l)}(c)] \\
&= p_{\mathrm{m}}(D_k^{(l)}(c))\frac{1}{|\mathtt{encode}^{(l)}(D_k^{(l)}(c))|} \\
&= \frac{1}{2^l}\cdot\frac{p_{\mathrm{m}}(D_k^{(l)}(c))}{p_{\mathrm{d}}(D_k^{(l)}(c))}.
\end{aligned}$$

For each $m\in\mathcal{M}$, $|p_{\mathrm{d}}(m) - p_{\mathrm{m}}(m)| \le \mathrm{SD}(M_0^{(l)}, M_1)$. Let $d = \mathrm{SD}(M_0^{(l)}, M_1)$, thus

$$\left|\frac{p_{\mathrm{m}}(m)}{p_{\mathrm{d}}(m)} - 1\right| \le \frac{d}{p_{\mathrm{d}}(m)}.$$

Let $p_{\mathrm{m}}^{\min} = \min_m p_{\mathrm{m}}(m)$, then $p_{\mathrm{d}}(m) \ge p_{\mathrm{m}}(m) - d \ge p_{\mathrm{m}}^{\min} - d$, further

$$\left|\frac{p_{\mathrm{m}}(m)}{p_{\mathrm{d}}(m)} - 1\right| \le \frac{d}{p_{\mathrm{m}}^{\min} - d}.$$

Thus,

$$\begin{aligned}
\left|p_{\mathrm{c|k}}(c\,|\,k) - \frac{1}{2^l}\right| &= \left|\frac{1}{2^l}\cdot\frac{p_{\mathrm{m}}(D_k^{(l)}(c))}{p_{\mathrm{d}}(D_k^{(l)}(c))} - \frac{1}{2^l}\right| \\
&\le \frac{1}{2^l}\cdot\frac{d}{p_{\mathrm{m}}^{\min} - d},
\end{aligned}$$

and

$$\begin{aligned}
\left|p_{\mathrm{c}}(c) - \frac{1}{2^l}\right| &= \left|\sum_{k\in\mathcal{K}} p_{\mathrm{k}}(k)p_{\mathrm{c|k}}(c\,|\,k) - \frac{1}{2^l}\right| \\
&\le \sum_{k\in\mathcal{K}} p_{\mathrm{k}}(k)\left|p_{\mathrm{c|k}}(c\,|\,k) - \frac{1}{2^l}\right| \\
&\le \frac{1}{2^l}\cdot\frac{d}{p_{\mathrm{m}}^{\min} - d}.
\end{aligned}$$

Denoting $U^{(l)}$ as the uniformly random variable on $\{0,1\}^l$, then

$$\begin{aligned}
\mathrm{SD}(C_{0,k}^{(l)}, U^{(l)}) &= \frac{1}{2}\sum_{c\in\mathcal{C}^{(l)}}\left|p_{\mathrm{c|k}}(c\,|\,k) - \frac{1}{2^l}\right| \\
&\le \frac{1}{2}\cdot 2^l\cdot\frac{1}{2^l}\cdot\frac{d}{p_{\mathrm{m}}^{\min} - d} = \frac{d}{2(p_{\mathrm{m}}^{\min} - d)},
\end{aligned}$$

and

$$\mathrm{SD}(C_1^{(l)}, U^{(l)}) \le \frac{d}{2(p_{\mathrm{m}}^{\min} - d)}.$$

This means the ciphertext $C_{0,k}^{(l)}$ and $C_0^{(l)}$ are both indistinguishable from $U^{(l)}$, and

$$\begin{aligned}
\mathrm{SD}(C_{0,k}^{(l)}, C_1^{(l)}) &\le \mathrm{SD}(C_{0,k}^{(l)}, U^{(l)}) + \mathrm{SD}(C_1^{(l)}, U^{(l)}) \\
&\le \frac{d}{p_{\mathrm{m}}^{\min} - d}
\end{aligned}$$

is negligible. By Theorem 1, the EE construction achieves KC.

However, the upper bound provided by Theorem 1 grows with the increase of $\frac{1}{p_{\mathrm{k}}^{\min}}$. We can give a tighter one for EE constructions which is independent of p_{k}.

Similar to $\frac{p_{\mathrm{m}}(m)}{p_{\mathrm{d}}(m)}$, we can get

$$\left|\frac{p_{\mathrm{d}}(m)}{p_{\mathrm{m}}(m)} - 1\right| \leq \frac{d}{p_{\mathrm{m}}(m)} \leq \frac{d}{p_{\mathrm{m}}^{\min}}.$$

Denoting $\epsilon_1 = \frac{d}{p_{\mathrm{m}}^{\min}}$ and $\epsilon_2 = \frac{d}{p_{\mathrm{m}}^{\min} - d}$, then for each m,

$$\frac{p_{\mathrm{d}}(m)}{p_{\mathrm{m}}(m)} \leq 1 + \epsilon_1, \text{ and } \frac{p_{\mathrm{m}}(m)}{p_{\mathrm{d}}(m)} \leq 1 + \epsilon_2.$$

Thus,

$$\frac{1}{2^l} \cdot \frac{1}{1+\epsilon_1} \leq p_{\mathrm{c|k}}(c|k) \leq \frac{1}{2^l} \cdot (1+\epsilon_2),$$

and

$$\frac{1}{2^l} \cdot \frac{1}{1+\epsilon_1} \leq p_{\mathrm{c}}(c) = \sum_{k \in \mathcal{K}} p_{\mathrm{k}}(k) p_{\mathrm{c|k}}(c|k) \leq \frac{1}{2^l} \cdot (1+\epsilon_2).$$

Further,

$$\frac{p_{\mathrm{c|k}}(c|k)}{p_{\mathrm{c}}(c)} \leq (1+\epsilon_1)(1+\epsilon_2),$$

and

$$\log \frac{p_{\mathrm{c|k}}(c|k)}{p_{\mathrm{c}}(c)} \leq \log(1+\epsilon_1) + \log(1+\epsilon_2) \leq \frac{\epsilon_1 + \epsilon_2}{\ln 2}.$$

Thus,

$$\begin{aligned} \mathrm{KL}(C_{0,k}^{(l)} || C_1^{(l)}) &= \sum_{c \in \mathcal{C}^{(l)}} p_{\mathrm{c|k}}(c|k) \log \frac{p_{\mathrm{c|k}}(c|k)}{p_{\mathrm{c}}(c)} \\ &\leq \frac{\epsilon_1 + \epsilon_2}{\ln 2}, \end{aligned}$$

and

$$\begin{aligned} I(K; C^{(l)}) &= \mathop{\mathbb{E}}_{K \leftarrow_{p_{\mathrm{k}}} \mathcal{K}} [\mathrm{KL}(C_{0,k}^{(l)} || C_1^{(l)})] \\ &\leq \frac{\epsilon_1 + \epsilon_2}{\ln 2} = \frac{d(2p_{\mathrm{m}}^{\min} - d)}{p_{\mathrm{m}}^{\min}(p_{\mathrm{m}}^{\min} - d) \ln 2} = O(d) \end{aligned}$$

is negligible. $\square$

Proposition 3 (restated). *For an EE construction, if its encoder is MS-IND, then $C_{0,k}^{(l)} \stackrel{s}{\equiv} C_1^{(l)} \stackrel{s}{\equiv} U_{k'}^{(l)}$, where $C_{0,k}^{(l)} \leftarrow_{p_{\mathrm{c|k}}(\cdot|k)} \mathcal{C}^{(l)}$, $C_1^{(l)} \leftarrow_{p_{\mathrm{c}}} \mathcal{C}^{(l)}$, and $U_{k'}^{(l)} \leftarrow_{\$} \mathcal{C}^{(l)} = \{0,1\}^l$. Specifically, with $l \geq \log 1/p_{\mathrm{m}}^{\min} + 1$, $M_0^{(l)} \leftarrow_{p_{\mathrm{d}}} \mathcal{M}$ and $M_1 \leftarrow_{p_{\mathrm{m}}} \mathcal{M}$,*

$$\mathrm{SD}(C_{0,k}^{(l)}, U^{(l)}) \leq \frac{\mathrm{SD}(M_0^{(l)}, M_1)}{2(p_{\mathrm{m}}^{\min} - \mathrm{SD}(M_0^{(l)}, M_1))}.$$

Proof of Proposition 3. From the proof of Theorem 3, it becomes evident that the ciphertext distribution is almost uniform. $\square$

Proposition 4 (restated). *For an EE construction, if for each m, $|p_{\mathrm{d}}(m) - p_{\mathrm{m}}(m)| \leq \epsilon$, where $0 < \epsilon < p_{\mathrm{m}}^{\min}$, then*

$$I(K; C^{(l)}) \leq \frac{\epsilon(2p_{\mathrm{m}}^{\min} - \epsilon)}{p_{\mathrm{m}}^{\min}(p_{\mathrm{m}}^{\min} - \epsilon) \ln 2} = O(\epsilon).$$

Proof of Proposition 4. The proof is similar to that of Theorem 3 which only uses $\mathrm{SD}(M_0, M_1)$ to bound $|p_{\mathrm{d}}(m) - p_{\mathrm{m}}(m)|$, for each $m \in \mathcal{M}$. $\square$

Theorem 4 (restated). *For n ciphertexts $C_1, C_2, \ldots, C_n$ encrypted with the same key K (possibly under different encryption schemes), if the messages are independent and the encryption randomness is independent, then*

$$I(K; C_1, C_2, \ldots, C_n) \leq \sum_{i=1}^{n} I(K; C_i).$$

Proof of Theorem 4.

$$\begin{aligned}
&I(K;C_1,C_2,\dots,C_n)\\
=&H(C_1,C_2,\dots,C_n)-H(C_1,C_2,\dots,C_n\,|\,K)\\
=&H(C_1,C_2,\dots,C_n)-\sum_i H(C_i\,|\,K)\\
\le&\sum_i H(C_i)-\sum_i H(C_i\,|\,K)\\
=&\sum_i I(K;C_i).
\end{aligned}$$

□

Theorem S.5. *In the ideal-cipher model where $\mathcal{C}=\mathcal{M}$ and each E_k is an independent uniformly random permutation over $\mathcal{M}$, the expected key leakage satisfies*

$$\mathbb{E}[I(K;C)]\ge\frac{|\mathcal{M}|\sum_m p_{\mathrm{m}}(m)^2-1}{2\ln 2\cdot|\mathcal{M}|\cdot p_{\mathrm{m}}^{\max}}\left(1-\sum_k p_{\mathrm{k}}(k)^2\right),$$

where the expectation is over the random choice of the ideal cipher.

Proof of Theorem S.5. In the ideal cipher model, the randomness arises from the fact that each encryption function E_k is an independently and uniformly chosen permutation over $\mathcal{M}$. For any fixed set of permutations $\{E_k\}_k$, the cipher becomes deterministic.

Using the equivalence relation R_{a}, defined in the proof of Theorem 2, we partition the ciphertext space $\mathcal{C}$ into $|\mathcal{M}|^{|\mathcal{K}|}$ equivalence classes. For each class $S_{\boldsymbol{m}}$ with $\boldsymbol{m}=(m_{j_i})_i$, we have $D_{k_i}(c)=m_{j_i}$ for each key k_i and each $c\in S_{\boldsymbol{m}}$.

Since the cipher is deterministic and $E_{k_i}(m_{j_i})=c$, we have:

$$\begin{aligned}
p_{\mathrm{c|k}}(c\,|\,k_i)&=p_{\mathrm{m}}(m_{j_i})p_{\mathrm{c|m,k}}(c\,|\,m_{j_i},k_i)\\
&=p_{\mathrm{m}}(m_{j_i}).
\end{aligned}$$

Thus $I(K;C)$ can be rewritten as:

$$\begin{aligned}
I(K;C)&=\sum_c\sum_i p_{\mathrm{k}}(k_i)p_{\mathrm{c|k}}(c\,|\,k_i)\log\frac{p_{\mathrm{c|k}}(c\,|\,k_i)}{p_{\mathrm{c}}(c)}\\
&=\sum_{\boldsymbol{m}}\sum_{c\in S_{\boldsymbol{m}}}\sum_i p_{\mathrm{k}}(k_i)p_{\mathrm{c|k}}(c\,|\,k_i)\log\frac{p_{\mathrm{c|k}}(c\,|\,k_i)}{\sum_{i'}p_{\mathrm{k}}(k_{i'})p_{\mathrm{c|k}}(c\,|\,k_{i'})}\\
&=\sum_{\boldsymbol{m}}|S_{\boldsymbol{m}}|\sum_i p_{\mathrm{k}}(k_i)p_{\mathrm{m}}(m_{j_i})\log\frac{p_{\mathrm{m}}(m_{j_i})}{\sum_{i'}p_{\mathrm{k}}(k_{i'})p_{\mathrm{m}}(m_{j_{i'}})}.
\end{aligned}$$

Since the randomness of the ideal cipher model only affects the value of $|S_{\boldsymbol{m}}|$, so

$$\begin{aligned}
\mathbb{E}[I(K;C)]=\sum_{\boldsymbol{m}}\mathbb{E}[|S_{\boldsymbol{m}}|]&\sum_i p_{\mathrm{k}}(k_i)p_{\mathrm{m}}(m_{j_i})\\
&\cdot\log\frac{p_{\mathrm{m}}(m_{j_i})}{\sum_{i'}p_{\mathrm{k}}(k_{i'})p_{\mathrm{m}}(m_{j_{i'}})}.
\end{aligned}$$

For the former part, noticed that $|S_{\boldsymbol{m}}|$ is binary, thus

$$\begin{aligned}
\mathbb{E}[|S_{\boldsymbol{m}}|]&=\Pr[|S_{\boldsymbol{m}}|=1]\\
&=\Pr[E_{k_i}(m_{j_i})=E_{k_1}(m^{(1)}),\forall i\ge 2]\\
&=\frac{1}{|\mathcal{M}|^{|\mathcal{K}|-1}}.
\end{aligned}$$

For the latter part, it can be transformed into

$$\sum_i p_{\mathrm{k}}(k_i)p_{\mathrm{m}}(m_{j_i})\log\frac{p_{\mathrm{m}}(m_{j_i})}{\sum_{i'}p_{\mathrm{k}}(k_{i'})p_{\mathrm{m}}(m_{j_{i'}})}$$

$$
\begin{aligned}
&= \sum_i p_{\mathrm{k}}(k_i) p_{\mathrm{m}}(m_{j_i}) \log p_{\mathrm{m}}(m_{j_i}) \\
&\quad - \sum_i p_{\mathrm{k}}(k_i) p_{\mathrm{m}}(m_{j_i}) \log \left(\sum_{i'} p_{\mathrm{k}}(k_{i'}) p_{\mathrm{m}}(m_{j_{i'}}) \right) \\
&= \sum_i \lambda_i f(x_i) - f(\overline{x}),
\end{aligned}
$$

where $\lambda_i = p_{\mathrm{k}}(k_i), x_i = p_{\mathrm{m}}(m_{j_i}), \overline{x} = \sum_i \lambda_i x_i, f(x) = x \log x$. According to Taylor's expansion,

$$
\begin{aligned}
\sum_i \lambda_i f(x_i) - f(\overline{x}) &= \sum_i \lambda_i \left(f(\overline{x}) + f'(\overline{x})(x_i - \overline{x}) + \frac{1}{2} f''(\xi_i)(x_i - \overline{x})^2 \right) - f(\overline{x}) \\
&= f(\overline{x}) + f'(\overline{x}) \sum_i \lambda_i (x_i - \overline{x}) + \frac{1}{2} \sum_i \lambda_i f''(\xi_i)(x_i - \overline{x})^2 - f(\overline{x}) \\
&= \frac{1}{2} \sum_i \lambda_i f''(\xi_i)(x_i - \overline{x})^2,
\end{aligned}
$$

where ξ_i lies between x_i and $\overline{x}$. It's clear that both of them can be upper-bounded by $p_{\mathrm{m}}^{\max}$. So $f''(\xi_i)$ has the following lower bound:

$$
f''(\xi_i) = \frac{1}{\xi_i \ln 2} \geq \frac{1}{p_{\mathrm{m}}^{\max} \ln 2}.
$$

Based on the above results, we have

$$
\mathbb{E}[I(K;C)] \geq \frac{\sum_{\boldsymbol{m}} \sum_i p_{\mathrm{k}}(k_i) \left(p_{\mathrm{m}}(m_{j_i}) - \sum_{i'} p_{\mathrm{k}}(k_{i'}) p_{\mathrm{m}}(m_{j_{i'}}) \right)^2}{2 \ln 2 |\mathcal{M}|^{|\mathcal{K}|-1} p_{\mathrm{m}}^{\max}}.
$$

For the numerator part,

$$
\begin{aligned}
&\sum_{\boldsymbol{m}} \sum_i p_{\mathrm{k}}(k_i) \left(p_{\mathrm{m}}(m_{j_i}) - \sum_{i'} p_{\mathrm{k}}(k_{i'}) p_{\mathrm{m}}(m_{j_{i'}}) \right)^2 \\
=& \sum_{\boldsymbol{m}} \left(\sum_i p_{\mathrm{k}}(k_i) p_{\mathrm{m}}(m_{j_i})^2 - \left(\sum_i p_{\mathrm{k}}(k_i) p_{\mathrm{m}}(m_{j_i}) \right)^2 \right) \\
=& \sum_{\boldsymbol{m}} \left(\sum_i \left(p_{\mathrm{k}}(k_i) - p_{\mathrm{k}}(k_i)^2 \right) p_{\mathrm{m}}(m_{j_i})^2 \right. \\
&\left. - \sum_{i \neq i'} p_{\mathrm{k}}(k_i) p_{\mathrm{k}}(k_{i'}) p_{\mathrm{m}}(m_{j_i}) p_{\mathrm{m}}(m_{j_{i'}}) \right) \\
=& \sum_i \left(p_{\mathrm{k}}(k_i) - p_{\mathrm{k}}(k_i)^2 \right) \sum_{\boldsymbol{m}} p_{\mathrm{m}}(m_{j_i})^2 \\
& - \sum_{i \neq i'} p_{\mathrm{k}}(k_i) p_{\mathrm{k}}(k_{i'}) \sum_{\boldsymbol{m}} p_{\mathrm{m}}(m_{j_i}) p_{\mathrm{m}}(m_{j_{i'}}) \\
=& \left(1 - \sum_i p_{\mathrm{k}}(k_i)^2 \right) |\mathcal{M}|^{|\mathcal{K}|-1} \sum_m p_{\mathrm{m}}(m)^2 \\
& - \left(1 - \sum_i p_{\mathrm{k}}(k_i)^2 \right) |\mathcal{M}|^{|\mathcal{K}|-2} \left(\sum_m p_{\mathrm{m}}(m) \right)^2 \\
=& |\mathcal{M}|^{|\mathcal{K}|-2} \left(1 - \sum_i p_{\mathrm{k}}(k_i)^2 \right) \left(|\mathcal{M}| \sum_m p_{\mathrm{m}}(m)^2 - 1 \right).
\end{aligned}
$$

Therefore,

$$
\mathbb{E}[I(K;C)] \geq \frac{|\mathcal{M}| \sum_m p_{\mathrm{m}}(m)^2 - 1}{2 \ln 2 |\mathcal{M}| p_{\mathrm{m}}^{\max}} \left(1 - \sum_k p_{\mathrm{k}}(k)^2 \right).
$$

$\square$

S.4. Counterexamples to the converses of Theorems 2 and 3

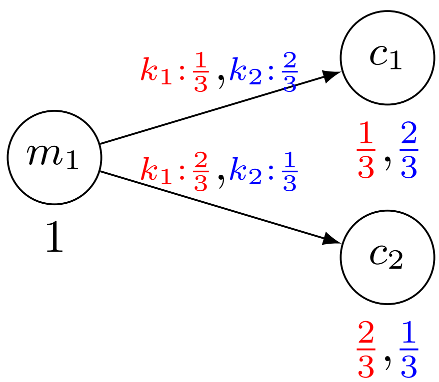

Figure S.2: Counterexample to the converse of Theorem 2

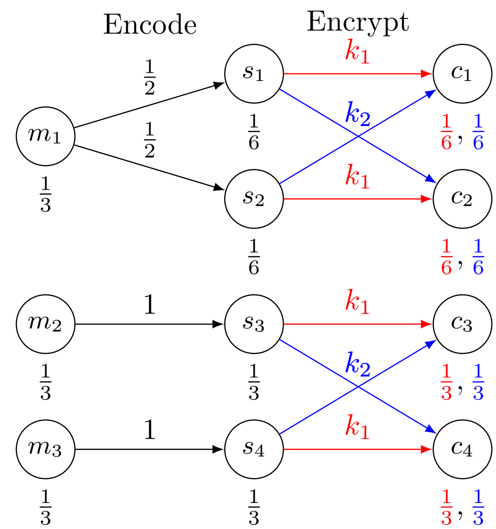

Figure S.3: Counterexample to the converse of Theorem 3: $p_{\mathrm{m}} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, and $p_{\mathrm{d}} = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$

The converse of Proposition 1 does not hold in general. As illustrated in Fig. S.2, a simple counterexample arises when, for the same message, E_k assigns different probabilities to different ciphertexts, resulting in distinct ciphertext distributions.

The converse of Theorem 3 does not hold. For certain ciphers with small key sizes, the ciphertext distribution $p_{\mathrm{c|k}}(\cdot\,|\,k)$ may remain consistent across different keys k, even when there are significant differences between p_{d} and p_{m}. Fig. S.3 presents a trivial counterexample, which naturally leads to more complex ones.

S.5. Implications of Mutual Information for Guessing-Based Security Notions

n-**GS Game** $\mathrm{GS}^{\mathcal{A}_n}_{p_{\mathrm{x,y}}}$
$X, Y \leftarrow_{p_{\mathrm{x,y}}} \mathcal{X} \times \mathcal{Y}$
$S \leftarrow \mathcal{A}_n(Y)$, where $\lvert S \rvert = n$
return $X \in S$

Figure S.4: n-Guessing Security Game (GS): guessing X given Y with n attempts

Security notions defined via mutual information inherently bound guessing-based security. We formalize this connection for general random variables X and Y, rather than restricting to specific cryptographic instantiations such as keys and ciphertexts.

Consider the n-guessing security game $\mathrm{GS}^{\mathcal{A}_n}_{p_{\mathrm{x,y}}}$ in Fig. S.4. Given Y, the adversary $\mathcal{A}_n$ outputs a set S of n candidate values for X. The advantage of $\mathcal{A}_n$ is defined as

$$\mathbf{Adv}^{n\text{-gs}}_{p_{\mathrm{x,y}}}(\mathcal{A}_n) = \Pr\left[\mathrm{GS}^{\mathcal{A}_n}_{p_{\mathrm{x,y}}} \Rightarrow \mathsf{True}\right],$$

and the maximum advantage is

$$\mathbf{Adv}^{n\text{-gs}}_{p_{\mathrm{x,y}}} = \max_{\mathcal{A}_n} \mathbf{Adv}^{n\text{-gs}}_{p_{\mathrm{x,y}}}(\mathcal{A}_n).$$

The optimal strategy of $\mathcal{A}_n$ is to output the n most likely values under the posterior distribution $p_{x|y}(\cdot\,|\,y)$. Let $S_{\mathrm{o}}(y, n)$ denote this optimal set for a given y, and let $S_{\mathrm{o}}(n)$ denote the set of n most likely values under the prior $p_x(\cdot)$.

Define

$$\mathrm{Guess}_n(X\,|\,Y) = \Pr[X \in S_{\mathrm{o}}(Y, n)], \quad \mathrm{Guess}_n(X) = \Pr[X \in S_{\mathrm{o}}(n)].$$

The following theorem quantifies the gain from observing Y.

Theorem S.6. *For any random variables X, Y and any integer $n \geq 1$,*

$$\mathbf{Adv}^{n\text{-}gs}_{p_{x,y}} = \mathrm{Guess}_n(X\,|\,Y) \leq \mathrm{Guess}_n(X) + \sqrt{\frac{\ln 2}{2} I(X;Y)}.$$

Proof. By Pinsker's inequality, for any y,

$$\mathrm{KL}\big(p_{x|y}(\cdot\,|\,y) \,\|\, p_x(\cdot)\big) \geq \frac{2}{\ln 2}\,\mathrm{SD}\big(p_{x|y}(\cdot\,|\,y), p_x(\cdot)\big)^2 .$$

Hence,

$$\mathrm{SD}\big(p_{x|y}(\cdot\,|\,y), p_x(\cdot)\big) \leq \sqrt{\frac{\ln 2}{2}\,\mathrm{KL}\big(p_{x|y}(\cdot\,|\,y) \,\|\, p_x(\cdot)\big)}.$$

Moreover, by the definition of statistical distance,

$$\sum_{x \in S_{\mathrm{o}}(y,n)} \big(p_{x|y}(x\,|\,y) - p_x(x)\big) \leq \mathrm{SD}\big(p_{x|y}(\cdot\,|\,y), p_x(\cdot)\big) .$$

Combining the above, we obtain

$$\begin{aligned}
&\mathrm{Guess}_n(X\,|\,Y) \\
=&\sum_y p_y(y) \sum_{x \in S_{\mathrm{o}}(y,n)} p_{x|y}(x\,|\,y) \\
\leq&\sum_y p_y(y) \sum_{x \in S_{\mathrm{o}}(y,n)} p_x(x) + \sum_y p_y(y)\sqrt{\frac{\ln 2}{2}\mathrm{KL}\big(p_{x|y}(\cdot\,|\,y) \,\|\, p_x(\cdot)\big)} \\
\leq&\sum_{x \in S_{\mathrm{o}}(n)} p_x(x) + \sqrt{\frac{\ln 2}{2}\sum_y p_y(y)\mathrm{KL}\big(p_{x|y}(\cdot\,|\,y) \,\|\, p_x(\cdot)\big)} \\
=&\mathrm{Guess}_n(X) + \sqrt{\frac{\ln 2}{2} I(X;Y)},
\end{aligned}$$

where the second inequality follows from Jensen's inequality. □

We next extend the result to partial-information recovery. Many guessing-based notions (e.g., TDSS) aim to recover a function of X rather than X itself.

Corollary S.3. *For any random variables X, Y and any function $f : \mathcal{X} \to \mathcal{Z}$, let $\omega_f = \max_z \Pr[f(X) = z]$. Then*

$$\mathrm{Guess}_1(f(X)\,|\,Y) \leq \omega_f + \sqrt{\frac{\ln 2}{2} I(X;Y)}.$$

Proof. Applying Theorem S.6 with $n = 1$ and replacing X by $f(X)$ gives

$$\mathrm{Guess}_1(f(X)\,|\,Y) \leq \omega_f + \sqrt{\frac{\ln 2}{2} I(f(X);Y)}.$$

Since $Y \to X \to f(X)$ forms a Markov chain, the data processing inequality implies $I(f(X);Y) \leq I(X;Y)$. □

Theorem S.6 and Corollary S.3 together show that statistical key confidentiality, expressed via mutual information, automatically bounds all guessing-based key recovery attacks, including those targeting arbitrary partial information and allowing multiple guesses.

S.6. Our OE_{U} for Uniform Key or Message Distributions

For simplicity, we denote the cumulative distribution functions of keys and messages as F_{k} and F_{m}, respectively. Thus, $F_{\mathrm{k}}(j) = \sum_{j'=1}^{j} p_{\mathrm{k}}(k_{j'})$ and $F_{\mathrm{m}}(i) = \sum_{i'=1}^{j} p_{\mathrm{m}}(m_{i'})$. Formally, OE_{U} has the following ciphertext intervals on $[0, 1)$. The corresponding discretized intervals on $\{0, 1\}^l$ can be naturally obtained.

In Case K: for a uniform key distribution,

$$\mathcal{C}(m_i, k_j) = [F_{\mathrm{m}}(i-1) + (j-1)F_{\mathrm{m}}(1), F_{\mathrm{m}}(i) + (j-1)F_{\mathrm{m}}(1)) ;$$

for a uniform message distribution,

$$\mathcal{C}(m_i,k_j)=\left[\frac{F_{\mathrm{m}}(1)}{F_{\mathrm{k}}(1)}F_{\mathrm{k}}(j-1)+(i-1)F_{\mathrm{m}}(1),\frac{F_{\mathrm{m}}(1)}{F_{\mathrm{k}}(1)}F_{\mathrm{k}}(j)+(i-1)F_{\mathrm{m}}(1)\right).$$

In Case M: for a uniform key distribution,

$$\mathcal{C}(m_i,k_j)=\left[\frac{F_{\mathrm{k}}(1)}{F_{\mathrm{m}}(1)}F_{\mathrm{m}}(i-1)+(j-1)F_{\mathrm{k}}(1),\frac{F_{\mathrm{k}}(1)}{F_{\mathrm{m}}(1)}F_{\mathrm{m}}(i)+(j-1)F_{\mathrm{k}}(1)\right);$$

for a uniform message distribution,

$$\mathcal{C}(m_i,k_j)=[F_{\mathrm{k}}(j-1)+(i-1)F_{\mathrm{k}}(1),F_{\mathrm{k}}(j)+(i-1)F_{\mathrm{k}}(1)).$$

S.7. Proofs in Section 5

Lemma 1 (restated). *In our design, $\mathcal{C}^{(l)}(m,k)\cap\mathcal{C}^{(l)}(m',k)=\emptyset$, for each $m\neq m'$ and k.*

Proof of Lemma 1 for Case K. In our assignment, $[l(m_i,k_i),u(m_i,k_i))$ for $i=1,2,\ldots,N_{\mathrm{m}}$ are non-overlapping intervals on $\mathbb{Z}$. Should any overlap occur within $\mathcal{C}^{(l)}$, meaning $\mathcal{C}^{(l)}(m_i,k)\cap\mathcal{C}^{(l)}(m_{i'},k_j)\neq\emptyset$ for some $i<i'$, it implies that $u(m_{i'},k_j)>l(m_i,k_j)+N_{\mathrm{c}}$. Consequently, this would suggest $u(m_{N_{\mathrm{m}}},k_j)>l(m_1,k_j)+N_{\mathrm{c}}$. However, our method ensures that $u(m_{N_{\mathrm{m}}},k_j)\leq l(m_1,k_j)+N_{\mathrm{c}}$, affirming the validity of the lemma.

In the following, we prove $u(m_{N_{\mathrm{m}}},k_j)\leq l(m_1,k_j)+N_{\mathrm{c}}$ for each j, recursively.

For $j=1$: the proposition obviously holds.

For $j>1$: Assume the proposition holds for $j-1$. Let $i_0=\max\{i\mid l(m_i,k_j)=u(m_i,k_{j-1})\}$. Note that such i exists, since $i=1$ meets the condition. Thus, $l(m_i,k_j)=u(m_{i-1},k_j)=l(m_{i-1},k_j)+n(m_{i-1})\frac{p_{\mathrm{k}}(k_j)}{p_{\mathrm{k}}(k_1)}$, for $i=i_0+1,...,N_{\mathrm{m}}$. Summarizing $l(m_i,k_j)-l(m_{i-1},k_j)$ over i yields

$$l(m_{N_M},k_j)-l(m_{i_0},k_j)=\sum_{i=i_0+1}^{N_{\mathrm{m}}}\lfloor n(m_{i-1})\frac{p_{\mathrm{k}}(k_j)}{p_{\mathrm{k}}(k_1)}\rfloor.$$

Moreover,

$$\begin{aligned}&u(m_{N_{\mathrm{m}}},k_j)-l(m_{i_0},k_j)\\=&\sum_{i=i_0}^{N_{\mathrm{m}}}\lfloor n(m_i)\frac{p_{\mathrm{k}}(k_j)}{p_{\mathrm{k}}(k_1)}\rfloor\leq\sum_{i=i_0}^{N_{\mathrm{m}}}\lfloor n(m_i)\frac{p_{\mathrm{k}}(k_{j-1})}{p_{\mathrm{k}}(k_1)}\rfloor\\\leq&\lfloor n(m_1)\frac{p_{\mathrm{k}}(k_{j-1})}{p_{\mathrm{k}}(k_1)}\rfloor+\sum_{i=i_0+1}^{N_{\mathrm{m}}}\lfloor n(m_i)\frac{p_{\mathrm{k}}(k_{j-1})}{p_{\mathrm{k}}(k_1)}\rfloor\\=&u(m_1,k_{j-1})-l(m_1,k_{j-1})+\sum_{i=i_0+1}^{N_M}\left(u(m_i,k_{j-1})-l(m_i,k_{j-1})\right).\end{aligned}$$

Applying $l(m_{i+1},k_j)=\max\{u(m_{i+1},k_{j-1}),u(m_i,k_j)\}\geq u(m_i,k_j)$ yields

$$\begin{aligned}&u(m_{N_{\mathrm{m}}},k_j)-l(m_{i_0},k_j)\\\leq&u(m_1,k_{j-1})-l(m_1,k_{j-1})+\sum_{i=i_0+1}^{N_M}\left(u(m_i,k_{j-1})-u(m_{i-1},k_{j-1})\right)\\=&u(m_1,k_{j-1})-l(m_1,k_{j-1})+u(m_{N_{\mathrm{m}}},k_{j-1})-u(m_{i_0},k_{j-1}).\end{aligned}$$

By the inductive hypothesis $u(m_{N_{\mathrm{m}}},k_{j-1})\leq l(m_1,k_{j-1})+N_{\mathrm{c}}$,

$$u(m_{N_{\mathrm{m}}},k_j)-l(m_{i_0},k_j)\leq u(m_1,k_{j-1})+N_{\mathrm{c}}-u(m_{i_0},k_{j-1}).$$

Further, since $l(m_{i_0},k_j)=u(m_{i_0},k_{j-1})$,

$$u(m_{N_{\mathrm{m}}},k_j)\leq u(m_1,k_{j-1})+N_{\mathrm{c}}=l(m_1,k_j)+N_{\mathrm{c}}.$$

Namely, the proposition holds for j.

The proposition implies that the intervals $[l(m_i,k_j),u(m_i,k_j))$ for $i=1,2,\dots,N_\mathrm{m}$ are non-overlapping on the space $[l(m_1,k_j),l(m_1,k_j)+N_\mathrm{c})$. Therefore, the corresponding intervals on $\mathcal{C}$ are also non-overlapping. □

Lemma 2 (restated). *In our design, $\mathcal{C}^{(l)}(m,k)\cap\mathcal{C}^{(l)}(m,k')=\emptyset$, for each m and $k\neq k'$.*

Proof of Lemma 2 for Case K. Similar to the proof of Lemma 1, we only need to prove that $u(m_i,k_{N_\mathrm{k}})\le l(m_i,k_1)+N_\mathrm{c}$ for each i.

In the following, we prove this recursively.

For $i=1$: for $j=1,2,...,N_\mathrm{k}-1$,

$$\begin{aligned}u(m_1,k_{j+1})&=l(m_1,k_{j+1})+\lfloor n(m_1)\frac{p_\mathrm{k}(k_{j+1})}{p_\mathrm{k}(k_1)}\rfloor\\&\le u(m_1,k_j)+n(m_1)\frac{p_\mathrm{k}(k_{j+1})}{p_\mathrm{k}(k_1)}.\end{aligned}$$

Iterating over j gets

$$\begin{aligned}u(m_1,k_{N_\mathrm{k}})&\le u(m_1,k_1)+\sum_{j=2}^{N_\mathrm{k}}n(m_1)\frac{p_\mathrm{k}(k_j)}{p_\mathrm{k}(k_1)}=l(m_1,k_1)+\sum_{j=1}^{N_\mathrm{k}}n(m_1)\frac{p_\mathrm{k}(k_j)}{p_\mathrm{k}(k_1)}\\&=l(m_1,k_1)+\frac{n(m_1)}{p_\mathrm{k}(k_1)}\le l(m_1,k_1)+N_\mathrm{c}.\end{aligned}$$

Namely, the proposition holds for $i=1$.

For $i>1$: Assume the proposition holds for $i-1$, i.e., $u(m_{i-1},k_{N_\mathrm{k}})\le l(m_{i-1},k_1)+N_\mathrm{c}$. Let $j_0=\max\{j\,|\,l(m_i,k_j)=u(m_i,k_{j-1})\}$. Note such j exists, because $j=1$ satisfies the condition. Then,

$$\begin{aligned}l(m_i,k_j)&=u(m_i,k_{j-1})\\&=l(m_i,k_{j-1})+n(m_i)\frac{p_\mathrm{k}(k_{j-1})}{p_\mathrm{k}(k_1)},\quad j=j_0+1,\dots,N_\mathrm{k}.\end{aligned}$$

Iterating over j yields

$$l(m_i,k_{N_\mathrm{k}})=l(m_i,k_{j_0})+\sum_{j=j_0+1}^{N_\mathrm{k}}\lfloor n(m_i)\frac{p_\mathrm{k}(k_{j-1})}{p_\mathrm{k}(k_1)}\rfloor.$$

Further,

$$\begin{aligned}&u(m_i,k_{N_\mathrm{k}})-l(m_i,k_{j_0})\\=&\sum_{j=j_0}^{N_\mathrm{k}}\lfloor n(m_i)\frac{p_\mathrm{k}(k_j)}{p_\mathrm{k}(k_1)}\rfloor\le\sum_{j=j_0}^{N_\mathrm{k}}\lfloor n(m_{i-1})\frac{p_\mathrm{k}(k_j)}{p_\mathrm{k}(k_1)}\rfloor\\\le&\lfloor n(m_{i-1})\frac{p_\mathrm{k}(k_1)}{p_\mathrm{k}(k_1)}\rfloor+\sum_{j=j_0+1}^{N_\mathrm{k}}\lfloor n(m_{i-1})\frac{p_\mathrm{k}(k_j)}{p_\mathrm{k}(k_1)}\rfloor\\=&u(m_{i-1},k_1)-l(m_{i-1},k_1)+\sum_{j=j_0+1}^{N_\mathrm{k}}\left(u(m_{i-1},k_j)-l(m_{i-1},k_j)\right).\end{aligned}$$

By $l(m_i,k_{j+1})=\max\{u(m_i,k_j),u(m_{i-1},k_{j+1})\}\ge u(m_i,k_j)$,

$$\begin{aligned}&u(m_i,k_{N_\mathrm{k}})-l(m_i,k_{j_0})\\\le&u(m_{i-1},k_1)-l(m_{i-1},k_1)+\sum_{j=j_0+1}^{N_\mathrm{k}}\left(u(m_{i-1},k_j)-u(m_{i-1},k_{j-1})\right)\\=&u(m_{i-1},k_1)-l(m_{i-1},k_1)+u(m_{i-1},k_{N_\mathrm{k}})-u(m_{i-1},k_{j_0}).\end{aligned}$$

With the inductive assumption $u(m_{i-1},k_{N_\mathrm{k}})\le l(m_{i-1},k_1)+N_\mathrm{c}$,

$$u(m_i,k_{N_\mathrm{k}})-l(m_i,k_{j_0})\le u(m_{i-1},k_1)+N_\mathrm{c}-u(m_{i-1},k_{j_0})$$

Further, since $l(m_i,k_{j_0})=u(m_i,k_{j_0-1})$, we have

$$u(m_i,k_{N_\mathrm{k}})\le u(m_{i-1},k_1)+N_\mathrm{c}=l(m_i,k_1)+N_\mathrm{c}.$$

Namely, the proposition holds for i.

The proposition naturally implies that the intervals $\mathcal{C}^{(l)}(m_i, k_j) = [l(m_i, k_j), u(m_i, k_j))$ for $j = 1, 2, \ldots, N_{\mathrm{k}}$ are non-overlapping on $\mathcal{C}^{(l)}$. In other words, the lemma holds. $\square$

Theorem 5 (restated). *For our scheme* OE*,*

$$\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{OE},p_{\mathrm{m}},p_{\mathrm{k}}} < p_{\max}\left(1 + \frac{t}{2^l - t}\right) = p_{\max} + O(2^{-l}),$$

where t is a constant $\frac{2p_{\max}}{p_{\mathrm{m}}^{\min}p_{\mathrm{k}}^{\min}}$.

Proof of Theorem 5. We prove the bound for Case K, where $p_{\max} = p_{\mathrm{k}}^{\max} = p_{\mathrm{k}}(k_1)$. The proof for Case M is symmetric by exchanging the roles of messages and keys.

As discussed before,

$$p_{\mathrm{m,c}}(m, c) = \sum_{k:D_k(c)=m} \frac{p_{\mathrm{m}}(m)p_{\mathrm{k}}(k)}{n(m, k)}.$$

Thus,

$$\begin{aligned}\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{OE},p_{\mathrm{m}},p_{\mathrm{k}}} &= \sum_{c\in\mathcal{C}} \max_m p_{\mathrm{m,c}}(m, c)\\ &= \sum_{c\in\mathcal{C}} \max_m \sum_{k:D_k(c)=m} \frac{p_{\mathrm{m}}(m)p_{\mathrm{k}}(k)}{n(m, k)}.\end{aligned}$$

Note that

$$\begin{aligned}n(m, k) &= \lfloor n(m)\frac{p_{\mathrm{k}}(k)}{p_{\mathrm{k}}(k_1)}\rfloor > n(m)\frac{p_{\mathrm{k}}(k)}{p_{\mathrm{k}}(k_1)} - 1\\ &> (p_{\mathrm{m}}(m)N_{\mathrm{c}} - 1)\frac{p_{\mathrm{k}}(k)}{p_{\mathrm{k}}(k_1)} - 1 > p_{\mathrm{m}}(m)N_{\mathrm{c}}\frac{p_{\mathrm{k}}(k)}{p_{\mathrm{k}}(k_1)} - 2.\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{p_{\mathrm{m}}(m)p_{\mathrm{k}}(k)}{n(m, k)} &< \frac{p_{\mathrm{m}}(m)p_{\mathrm{k}}(k)}{p_{\mathrm{m}}(m)N_{\mathrm{c}}\frac{p_{\mathrm{k}}(k)}{p_{\mathrm{k}}(k_1)} - 2} = \frac{p_{\mathrm{k}}(k_1)}{N_{\mathrm{c}} - \frac{2p_{\mathrm{k}}(k_1)}{p_{\mathrm{m}}(m)p_{\mathrm{k}}(k)}}\\ &\le \frac{p_{\mathrm{k}}(k_1)}{N_{\mathrm{c}} - \frac{2p_{\mathrm{k}}(k_1)}{p_{\mathrm{m}}(m_{N_{\mathrm{m}}})p_{\mathrm{k}}(k_{N_{\mathrm{k}}})}}.\end{aligned}$$

Further, by Lemmas 1 and 2, $|\{k \,|\, D_k(c) = m\}| \le 1$, and

$$\begin{aligned}\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{OE},p_{\mathrm{m}},p_{\mathrm{k}}} &< N_{\mathrm{c}} \cdot \frac{p_{\mathrm{k}}(k_1)}{N_{\mathrm{c}} - \frac{2p_{\mathrm{k}}(k_1)}{p_{\mathrm{m}}(m_{N_{\mathrm{m}}})p_{\mathrm{k}}(k_{N_{\mathrm{k}}})}}\\ &= p_{\mathrm{k}}(k_1) + \frac{\frac{2p_{\mathrm{k}}(k_1)^2}{p_{\mathrm{m}}(m_{N_{\mathrm{m}}})p_{\mathrm{k}}(k_{N_{\mathrm{k}}})}}{N_{\mathrm{c}} - \frac{2p_{\mathrm{k}}(k_1)}{p_{\mathrm{m}}(m_{N_{\mathrm{m}}})p_{\mathrm{k}}(k_{N_{\mathrm{k}}})}}.\end{aligned}$$

Note $N_{\mathrm{c}} = 2^l$, thus $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{OE},p_{\mathrm{m}},p_{\mathrm{k}}} = p_{\mathrm{k}}(k_1) + O\left(\frac{1}{2^l}\right)$. In Case M, the same argument with m and k interchanged gives $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{OE},p_{\mathrm{m}},p_{\mathrm{k}}} = p_{\mathrm{m}}(m_1) + O(2^{-l})$. Therefore, in both cases,

$$\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{OE},p_{\mathrm{m}},p_{\mathrm{k}}} < p_{\max}\left(1 + \frac{t}{2^l - t}\right) = p_{\max} + O(2^{-l}),$$

where $t = \frac{2p_{\max}}{p_{\mathrm{m}}^{\min}p_{\mathrm{k}}^{\min}}$ is a constant. $\square$

Theorem 6 (restated). *For the scheme OE' constructed via piecewise uniform approximation with parameter e^{ϵ}, we have*

$$\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{OE}',p_{\mathrm{m}},p_{\mathrm{k}}} \le e^{3\epsilon}p_{\max} + O(2^{-l}),$$

with computational complexity $O(n_{\mathrm{KBin}} \cdot n_{\mathrm{MBin}})$, where n_{KBin} and n_{MBin} are the numbers of key and message bins, respectively.

Proof of Theorem 6. Using the continuous variants OE_{c} and $\mathsf{OE}'_{\mathrm{c}}$, we have

$$|\mathcal{C}(m,k)| = \frac{p_{\mathrm{m}}(m)p_{\mathrm{k}}(k)}{p_{\max}}, \quad |\mathcal{C}'(m,k)| = \frac{p'_{\mathrm{m}}(m)p'_{\mathrm{k}}(k)}{p'_{\max}}.$$

By construction, for all m and k, the piecewise uniform approximations satisfy

$$e^{-\epsilon} \le \frac{p'_{\mathrm{m}}(m)}{p_{\mathrm{m}}(m)} \le e^{\epsilon}, \quad e^{-\epsilon} \le \frac{p'_{\mathrm{k}}(k)}{p_{\mathrm{k}}(k)} \le e^{\epsilon}, \quad e^{-\epsilon} \le \frac{p'_{\max}}{p_{\max}} \le e^{\epsilon}.$$

Therefore, the ratio $\frac{|\mathcal{C}(m,k)|}{|\mathcal{C}'(m,k)|} \le e^{3\epsilon}$.

Then, under p_{m} and p_{k}, it holds that

$$\begin{aligned}
\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{OE}'_{\mathrm{c}},p_{\mathrm{m}},p_{\mathrm{k}}} &= \int_c \max_m p'_{\mathrm{m,c}}(m,c)\,dc \\
&= \int_c \max_m \frac{p_{\mathrm{m}}(m)p_{\mathrm{k}}(k)}{|\mathcal{C}'(m,k)|}\,dc \\
&\le \int_c \max_m \frac{p_{\mathrm{m}}(m)p_{\mathrm{k}}(k)}{|\mathcal{C}(m,k)|} e^{3\epsilon}\,dc \\
&= e^{3\epsilon} \cdot \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{OE}_{\mathrm{c}},p_{\mathrm{m}},p_{\mathrm{k}}} = e^{3\epsilon} p_{\max}.
\end{aligned}$$

Hence,

$$\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{OE}',p_{\mathrm{m}},p_{\mathrm{k}}} \le e^{3\epsilon} p_{\max} + O(2^{-l}).$$

Within each bin, ciphertext intervals have the same size for all messages and keys, and their endpoints can be computed as in OE_{U}, requiring only $O(1)$ time. Across the entire space, OE' only needs to store $l(m,k)$ and $n(m,k)$ for each representative (m,k) pair at the beginning of each bin. The ciphertext intervals for other (m,k) pairs can then be derived in $O(1)$ time using these values. Therefore, the overall complexity of OE' is $O(n_{\mathrm{KBin}} \cdot n_{\mathrm{MBin}})$. ☐

Proposition 6 (restated). *For $\mathcal{M} = \{a,b\}$ with $p_{\mathrm{m}} = (\frac{1}{2}, \frac{1}{2})$ and any $\mathcal{K}$ such that $p_{\mathrm{k}}^{\max} \le \frac{1}{2}$ and for any subsets $\mathcal{K}_1 \ne \mathcal{K}_2$ with $p_{\mathrm{k}}(\mathcal{K}_1) \ne p_{\mathrm{k}}(\mathcal{K}_2)$, it holds that*

$$\inf_{\mathsf{EE} \in \Omega_{EE}} \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}} > p_{\max} = \tfrac{1}{2}.$$

Proof of Proposition 6. Let $n_m = n(m,k)$ as in EE constructions, $n(m,k)$ is independent of k. Without loss of generality, assume that $n_a \ge n_b$. Then

$$\begin{aligned}
\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}} &= \sum_{c \in \mathcal{C}} \max_m \left\{ \sum_{k:D_k(c)=m} \frac{p_{\mathrm{k}}(k)p_{\mathrm{m}}(m)}{n(m,k)} \right\} \\
&= \sum_{c \in \mathcal{C}} \max \left\{ \sum_{k:D_k(c)=a} \frac{p_{\mathrm{k}}(k)p_{\mathrm{m}}(a)}{n(a,k)}, \sum_{k:D_k(c)=b} \frac{p_{\mathrm{k}}(k)p_{\mathrm{m}}(b)}{n(b,k)} \right\} \\
&= \sum_{c \in \mathcal{C}} \max \left\{ \frac{p_{\mathrm{m}}(a)}{n_a} \sum_{k:D_k(c)=a} p_{\mathrm{k}}(k), \frac{p_{\mathrm{m}}(b)}{n_b} \sum_{k:D_k(c)=b} p_{\mathrm{k}}(k) \right\}.
\end{aligned}$$

Note that each term of the summation is related to its decryption result under each key. Define $f : \mathcal{C} \to \mathcal{M}^{N_{\mathrm{k}}}$ such that $f(c) = (D_{k_i}(c))_{i=1}^{N_{\mathrm{k}}}$. Using f, the ciphertext space is partitioned into distinct equivalence classes $\mathcal{C}_x$ that for each $x \in \mathcal{M}^{N_{\mathrm{k}}}$, $\mathcal{C}_x = \{c \,|\, f(c) = x\}$. Further, by $g(x) = \sum_{i:x_i=a} p_{\mathrm{k}}(k_i)$, we have

$$\begin{aligned}
\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}} &= \sum_{c \in \mathcal{C}} \max \left\{ \frac{p_{\mathrm{m}}(a)}{n_a} \sum_{k:D_k(c)=a} p_{\mathrm{k}}(k) \frac{p_{\mathrm{m}}(b)}{n_b} \sum_{k:D_k(c)=b} p_{\mathrm{k}}(k) \right\} \\
&= \sum_{x \in \mathcal{M}^{N_{\mathrm{k}}}} \sum_{c \in \mathcal{C}_x} \max \left\{ \frac{p_{\mathrm{m}}(a)}{n_a} \sum_{k:x_i=a} p_{\mathrm{k}}(k) \frac{p_{\mathrm{m}}(b)}{n_b} \sum_{k:x_i=b} p_{\mathrm{k}}(k) \right\} \\
&= \sum_{x \in \mathcal{M}^{N_{\mathrm{k}}}} |\mathcal{C}_x| \cdot \max \left\{ \frac{p_{\mathrm{m}}(a)}{n_a} g(x), \frac{p_{\mathrm{m}}(b)}{n_b} (1 - g(x)) \right\}
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{x\in\mathcal{M}^{N_\mathrm{k}}} |\mathcal{C}_x|\cdot\left(\frac{p_\mathrm{m}(a)}{n_a}g(x)+\frac{p_\mathrm{m}(b)}{n_b}(1-g(x))+\left|\frac{p_\mathrm{m}(a)}{n_a}g(x)-\frac{p_\mathrm{m}(b)}{n_b}(1-g(x))\right|\right)\\
&= \frac{1}{2}+\frac{1}{2}\sum_{x\in\mathcal{M}^{N_\mathrm{k}}} |\mathcal{C}_x|\cdot\left|\left(\frac{p_\mathrm{m}(a)}{n_a}+\frac{p_\mathrm{m}(b)}{n_b}\right)g(x)-\frac{p_\mathrm{m}(b)}{n_b}\right|\\
&= \frac{1}{2}+\frac{1}{2}\sum_{x\in\mathcal{M}^{N_\mathrm{k}}} |\mathcal{C}_x|\cdot\left|(\alpha+\beta)\,g(x)-\beta\right|,
\end{aligned}
$$

here $\alpha=\frac{p_\mathrm{m}(a)}{n_a}=\frac{1}{2n_a}$, $\beta=\frac{p_\mathrm{m}(b)}{n_b}=\frac{1}{2n_b}$. Notice that $\alpha\le\beta$ and

$$
\alpha+\beta=\frac{1}{2n_a}+\frac{1}{2n_b}\ge\frac{2}{n_a+n_b}=\frac{2}{N_\mathrm{c}}.
$$

Denote $\mathcal{T}=\{g(x)\mid x\in\mathcal{M}^{N_\mathrm{k}}\}$. Since $p_\mathrm{k}(\mathcal{K}_1)\neq p_\mathrm{k}(\mathcal{K}_2)$ for any subsets $\mathcal{K}_1\neq\mathcal{K}_2$, $\mathcal{T}$ consists of $|\mathcal{M}^{N_\mathrm{k}}|$ distinct elements. Assume that the elements in $\mathcal{T}$ are arranged in ascending order as $\{t_1,t_2,...,t_{N_\mathrm{t}}\}$. Further, each $t_i\in\mathcal{T}$ corresponds to a unique element in $\mathcal{M}^{N_\mathrm{k}}$, denoted by x_i. As a simple observation, $x_0=(b,b,...,b)$ and $t_0=0$, while $x_{N_\mathrm{t}}=(a,a,...,a)$ and $t_{N_\mathrm{t}}=1$. Denoting $\delta=\min_{t,t'\in\mathcal{T}}|t-t'|$, we provide a lower bound of $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_\mathrm{m},p_\mathrm{k}}$ based on δ. We discuss the bound in two cases: 1) $n_a\le t_{N_\mathrm{t}-1}N_\mathrm{c}$ and 2) $n_a>t_{N_\mathrm{t}-1}N_\mathrm{c}$.

Case 1: $n_a\le t_{N_\mathrm{t}-1}N_\mathrm{c}$:

Define i by $t_i<n_a/N_\mathrm{c}\le t_{i+1}$, and let $\mathcal{S}=\mathcal{M}^{N_\mathrm{k}}\backslash\{x_i,x_{i+1}\}$. Since $\frac{\beta}{\alpha+\beta}=n_a/N_\mathrm{c}\in(t_i,t_{i+1}]$, we have $(\alpha+\beta)t_i<\beta$ and $(\alpha+\beta)t_{i+1}\ge\beta$

If $|\mathcal{C}_{x_i}|+|\mathcal{C}_{x_{i+1}}|<3N_\mathrm{c}/4$: $\sum_{x\in\mathcal{S}}|\mathcal{C}_x|\ge N_\mathrm{c}/4$. For $j>i+1$, $|(\alpha+\beta)g(x_j)-\beta|=(\alpha+\beta)t_j-\beta\ge(\alpha+\beta)(t_{i+1}+\delta)-\beta\ge(\alpha+\beta)\delta\ge2\delta/N_\mathrm{c}$. Similarly, for $j<i$, $|(\alpha+\beta)g(x_j)-\beta|=\beta-(\alpha+\beta)t_j\ge\beta-(\alpha+\beta)(t_i-\delta)\ge(\alpha+\beta)\delta\ge2\delta/N_\mathrm{c}$. Combining the inequalities together yields

$$
\begin{aligned}
\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_\mathrm{m},p_\mathrm{k}} &\ge \frac{1}{2}+\frac{1}{2}\sum_{x\in\mathcal{M}^{N_\mathrm{k}}}|\mathcal{C}_x|\cdot|(\alpha+\beta)\,g(x)-\beta|\\
&\ge \frac{1}{2}+\frac{1}{2}\sum_{x\in\mathcal{S}}|\mathcal{C}_x|\cdot|(\alpha+\beta)\,g(x)-\beta|\\
&\ge \frac{1}{2}+\frac{1}{2}\cdot\frac{2\delta}{N_\mathrm{c}}\sum_{x\in T}|\mathcal{C}_x|\\
&\ge \frac{1}{2}+\frac{\delta}{4}.
\end{aligned}
$$

If $|\mathcal{C}_{x_i}|+|\mathcal{C}_{x_{i+1}}|\ge3N_\mathrm{c}/4$: Since $t_i<n_a/N_\mathrm{c}\le t_{N_\mathrm{t}-1}$, we have $i<N_\mathrm{t}-1$. Besides, $t_{i+1}\ge n_a/N_\mathrm{c}\ge\frac{1}{2}>t_1$ implies $i>1$. Note that for $x\in\mathcal{M}^{N_\mathrm{k}}\backslash\{x_0,x_{N_\mathrm{t}}\}$, $|\mathcal{C}_x|\le N_\mathrm{c}/2$ (otherwise, there exist j,j' such that $x_j=a$ and $x_{j'}=b$, and further $n_a>|\mathcal{C}_x|>N_\mathrm{c}/2$ and $n_b>|\mathcal{C}_x|>N_\mathrm{c}/2$, leading to contradiction). Therefore, $|\mathcal{C}_{x_i}|\ge3N_\mathrm{c}/4-|\mathcal{C}_{x_{i+1}}|\ge N_\mathrm{c}/4$. Similarly, $|\mathcal{C}_{x_{i+1}}|\ge N_\mathrm{c}/4$. Thus,

$$
\begin{aligned}
\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_\mathrm{m},p_\mathrm{k}} &\ge \frac{1}{2}+\frac{1}{2}\sum_{x\in\mathcal{M}^{N_\mathrm{k}}}|\mathcal{C}_x|\cdot|(\alpha+\beta)\,g(x)-\beta|\\
&\ge \frac{1}{2}+\frac{1}{2}|\mathcal{C}_{x_i}|\cdot|(\alpha+\beta)\,g(x_i)-\beta|\\
&\quad+\frac{1}{2}|\mathcal{C}_{x_{i+1}}|\cdot|(\alpha+\beta)\,g(x_{i+1})-\beta|\\
&\ge \frac{1}{2}+\frac{N_\mathrm{c}}{8}\,|(\alpha+\beta)\,g(x_i)-\beta|\\
&\quad+\frac{N_\mathrm{c}}{8}\,|(\alpha+\beta)\,g(x_{i+1})-\beta|\\
&\ge \frac{1}{2}+\frac{N_\mathrm{c}}{8}\,|(\alpha+\beta)\,g(x_i)-(\alpha+\beta)\,g(x_{i+1})|\\
&\ge \frac{1}{2}+\frac{N_\mathrm{c}}{8}\cdot\delta(\alpha+\beta)\\
&\ge \frac{1}{2}+\frac{\delta}{4}.
\end{aligned}
$$

Case 2: $n_a>t_{N_\mathrm{t}-1}N_\mathrm{c}$:

Note $\alpha g(x) > \beta(1-g(x))$ only holds if $x = x_{N_\mathrm{t}}$. Therefore,

$$\begin{aligned}\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_\mathrm{m},p_\mathrm{k}} &= \sum_{x\in\mathcal{M}^{N_\mathrm{k}}} |\mathcal{C}_x| \cdot \max\{\alpha g(x), \beta(1-g(x))\} \\ &= \alpha|\mathcal{C}_{x_0}| + \sum_{x\in\mathcal{M}^{N_\mathrm{k}}\setminus\{x_0\}} |C_x| \cdot \beta(1-g(x)) \\ &= \alpha|\mathcal{C}_{x_0}| + \sum_{c\in\mathcal{C}}\sum_{k:D_k(c)=b} \beta \\ &= p_\mathrm{m}(b) + \alpha|\mathcal{C}_{x_0}| \\ &= \frac{1}{2} + \frac{|\mathcal{C}_{x_0}|}{2n_a}.\end{aligned}$$

On one hand, $\sum_{k\in\mathcal{K}}\sum_{c\in\mathcal{C}} 1_{D_k(c)=a} = \sum_{k\in\mathcal{K}} n_a = N_\mathrm{k} n_a$. On the other hand,

$$\begin{aligned}\sum_{c\in\mathcal{C}}\sum_{k\in\mathcal{K}} 1_{D_k(c)=a} &= \sum_{c\in\mathcal{C}_{x_0}}\sum_{k\in\mathcal{K}} 1_{D_k(c)=a} + \sum_{c\in\mathcal{C}\setminus\mathcal{C}_{x_0}}\sum_{k\in\mathcal{K}} 1_{D_k(c)=a} \\ &\le N_\mathrm{k}|\mathcal{C}_{x_0}| + (N_\mathrm{k}-1)(N_\mathrm{c} - |\mathcal{C}_{x_0}|) \\ &= (N_\mathrm{k}-1)N_\mathrm{c} + |\mathcal{C}_{x_0}| \\ &< (N_\mathrm{k}-1)\frac{n_a}{t_{N_\mathrm{t}-1}} + |\mathcal{C}_{x_0}| \\ &= \frac{n_a(N_\mathrm{k}-1)}{1-p_\mathrm{k}^{\min}} + |\mathcal{C}_{x_0}|.\end{aligned}$$

Thus, $\frac{|\mathcal{C}_{x_0}|}{n_a} > \frac{1-N_\mathrm{k}\cdot p_\mathrm{k}^{\min}}{1-p_\mathrm{k}^{\min}} > 0$. Let $\lambda = \frac{1-N_\mathrm{k}\cdot p_\mathrm{k}^{\min}}{1-p_\mathrm{k}^{\min}}$, we have

$$\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_\mathrm{m},p_\mathrm{k}} \ge \frac{1}{2} + \frac{\lambda}{2}.$$

With the bounds in the two cases,

$$\inf_{\mathsf{EE}\in\Omega_{\mathsf{EE}}} \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_\mathrm{m},p_\mathrm{k}} \ge \frac{1}{2} + \frac{1}{2}\min\{\frac{\delta}{2}, \lambda\} > \frac{1}{2}.$$

Namely, the theorem holds. □

Proposition 9 (restated). *If a probabilistic encryption scheme $(\mathcal{M},\mathcal{K},\mathcal{C},\mathcal{E},\mathcal{D})$ is uniform, message-dependent, and key-independent, then it can be represented as an EE construction $(\mathcal{M},\mathcal{K},\mathcal{C},\mathcal{E}',\mathcal{D}')$ such that for all $m\in\mathcal{M}$, $k\in\mathcal{K}$, and $c\in\mathcal{C}$,*

$$\Pr[E_k(m) = c] = \Pr[E'_k(m) = c].$$

Proof of Proposition 9. Since for each m, $|\mathcal{C}(m,k)|$ is independent of k, we can establish $p_\mathrm{d}(m) = \frac{|\mathcal{C}(m,k)|}{2^l}$ with an appropriate encoder $(\mathtt{encode}, \mathtt{decode})$. Further, we can design a cipher $(\mathtt{encrypt}, \mathtt{decrypt})$ on $\{0,1\}^l$ such that $\mathtt{encrypt}_k$ transforms the seeds in $\mathtt{encode}(m)$ to the ciphertext in $\mathcal{C}(m,k)$. In this way, we construct an EE construction that, under the key k, encrypts the message m to ciphertexts in $\mathcal{C}(m,k)$ with the same probability $\frac{1}{|\mathcal{C}(m,k)|}$. Therefore, for each $m\in\mathcal{M}$, $k\in\mathcal{K}$: if $c\in\mathcal{C}(m,k)$, then $\Pr[E_k(m)=c] = \frac{1}{|\mathcal{C}(m,k)|} = \Pr[E'_k(m)=c]$; if $c\notin\mathcal{C}(m,k)$, then $\Pr[E_k(m)=c] = 0 = \Pr[E'_k(m)=c]$. In other words, the EE construction satisfies the requirement in the theorem. □

Proposition 7 (restated). OE *achieves both optimal KC and optimal MRS if and only if the key distribution is uniform and $p_\mathrm{m}^{\max} \le \frac{1}{|\mathcal{K}|}$.*

Proof of Proposition 7. We demonstrate this by analyzing the following three cases:

1. When the key distribution is non-uniform, the ciphertext set size in OE satisfies

$$|\mathcal{C}^{(l)}(m,k)| \propto p_\mathrm{m}(m)p_\mathrm{k}(k) + O(2^{-l}),$$

which deviates from the requirement for optimal KC:

$$|\mathcal{C}^{(l)}(m,k)| \propto p_\mathrm{m}(m) + \mathsf{negl}(l).$$

TDSS Game $\mathrm{TDSS}^{\mathcal{A},f}_{\mathsf{E},p_{\mathrm{m}},p_{\mathrm{k}}}$
$M \leftarrow_{p_{\mathrm{m}}} \mathcal{M}$
$K \leftarrow_{p_{\mathrm{k}}} \mathcal{K}$
$C \leftarrow E_K(M)$
$b \leftarrow \mathcal{A}(C)$
return $b = f(M)$

Figure S.5: Game for TDSS security

2. If the key distribution is uniform but $p_{\mathrm{m}}(m_1) > \frac{1}{|\mathcal{K}|}$, the ciphertext support varies across different keys, violating the condition for optimal KC. More specifically,
$$|\mathcal{C}^{(l)}(m,k)| \approx 2^l \cdot p_{\mathrm{k}}(k) \cdot \frac{p_{\mathrm{m}}(m)}{p_{\mathrm{m}}(m_1)} = \frac{2^l p_{\mathrm{m}}(m)}{p_{\mathrm{m}}(m_1)|\mathcal{K}|},$$
$$\sum_m |\mathcal{C}^{(l)}(m,k)| \approx \frac{2^l}{p_{\mathrm{m}}(m_1)|\mathcal{K}|} < 2^l.$$
This implies that in order to prevent collisions on m_1 under different keys, OE must assign a non-empty $\mathcal{C}^{(l)}(\perp,k)$, which differs across keys. Consequently, the ciphertext distributions vary with the key, violating KC.
3. When the key distribution is uniform and $p_{\mathrm{m}}(m_1) \le \frac{1}{|\mathcal{K}|}$, we have $\mathcal{C}^{(l)}(\perp,k) = \emptyset$, and the ciphertext distributions across keys are nearly uniform on $\mathcal{C}^{(l)}$. Therefore, OE satisfies both optimal KC and MRS in this case.

□

S.8. Target-distribution Semantic Security

As a stronger notion of MC [11], target-distribution semantic-security (TDSS) models the leakage of partial information about the message from the ciphertext.

Formally, for a predicate $f : \mathcal{M} \to \{0,1\}$, the TDSS game is shown in Fig. S.5. An adversary $\mathcal{A}$ is going to predict $f(M)$ given the ciphertext C of M. Clearly, a trivial adversary can simply output the most likely value of $f(M)$ and achieves the success probability of $\omega_f = \max_b \Pr[f(M) = b]$. Considering this, Jaeger et al. [11] defined the TDSS advantage as:
$$\mathbf{Adv}^{\mathrm{tdss}}_{\mathsf{E},p_{\mathrm{m}},p_{\mathrm{k}}} = \max_{\mathcal{A},f} \left\{ \Pr[\mathrm{TDSS}^{\mathcal{A},f}_{\mathsf{E},p_{\mathrm{m}},p_{\mathrm{k}}} \Rightarrow \mathrm{true}] - \omega_f \right\}.$$

Subtracting ω_f aims to ensure that the minimum TDSS advantage is zero. However, we show that zero TDSS advantage is generally unattainable for arbitrary message and key distributions.

Theorem S.7. *If $|\mathcal{M}| = |\mathcal{K}| = 2$ and $p_{\mathrm{k}}(k_1) > p_{\mathrm{m}}(m_1)$, then for any symmetric encryption scheme E, $\mathbf{Adv}^{\mathrm{tdss}}_{\mathsf{E},p_{\mathrm{m}},p_{\mathrm{k}}} \ge p_{\mathrm{k}}(k_1) - p_{\mathrm{m}}(m_1) > 0$.*

Proof. Since $|\mathcal{M}| = 2$, it suffices to consider the trivial predicate $f(m_i) = i - 1$. In this case, the TDSS adversary must predict the exact message, making the TDSS game equivalent to the MRS game. Since the trivial MRS advantage is $p_{\mathrm{k}}(k_1)$, we have
$$\mathbf{Adv}^{\mathrm{tdss}}_{\mathsf{E},p_{\mathrm{m}},p_{\mathrm{k}}} = \max_{\mathcal{A}} \Pr[\mathrm{TDSS}^{\mathcal{A},f}_{\mathsf{E},p_{\mathrm{m}},p_{\mathrm{k}}} \Rightarrow 1] - \omega_f \ge p_{\mathrm{k}}(k_1) - p_{\mathrm{m}}(m_1) > 0.$$

□

We leave the study of the optimal attainable TDSS advantage under general message and key distributions as future work.

S.9. Details of Continuous Honey Encryption Schemes and Proofs

Security of uniformized encryption schemes. Intuitively, uniformizing $p_{\mathrm{c|m,k}}$ over each equivalence class S of R_{a} induces a new ciphertext C' from the original ciphertext C via a post-processing step: first identify the equivalence class S to which C belongs, and then sample C' uniformly from S. As a result, C' reveals only the equivalence class of C. Since C' is obtained by post-processing C, it cannot reveal more information about the underlying variables than C.

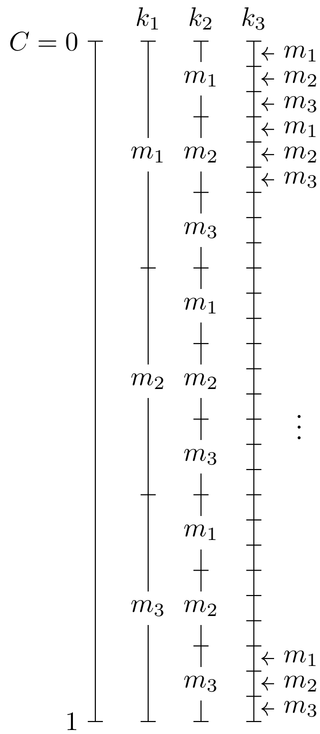

Figure S.6: Rearrangement of ciphertext intervals, where $m_i \in \mathcal{M}' = \mathcal{M} \cup \{\perp\}$

To formalize this intuition, we consider a general measure $M(X;Y)$ that quantifies the amount of information about a sensitive variable X revealed by an observation Y. Formally, M is a functional that maps the joint distribution $P_{X,Y}$ to a real value:

$$M : \mathcal{P}(\mathcal{X} \times \mathcal{Y}) \to \mathbb{R}.$$

Depending on the context, M may represent an information-theoretic quantity (e.g., mutual information or divergence) or an operational quantity (e.g., optimal guessing success probability).

Definition S.2 (Post-processing monotonicity)**.** *A measure $M(X;Y)$ is said to be monotone under post-processing if for any Markov chain $X \to Y \to Z$, it holds that*

$$M(X;Z) \leq M(X;Y).$$

This property is a standard abstraction of the data processing inequality (DPI) in information theory [35], and is closely related to Blackwell's comparison of statistical experiments [36]. A wide class of measures satisfy this property, including f-divergences [37, 38], Shannon mutual information, Rényi divergence and Sibson mutual information [39, 40], as well as decision-theoretic quantities such as optimal guessing success probability. More recently, maximal leakage has also been shown to satisfy post-processing monotonicity [41]. In this paper, we focus on mutual information and guessing success probability as security measures, both of which satisfy post-processing monotonicity.

Proposition S.3. *$\mathrm{Guess}_n(X\,|\,Y)$ and $I(X;Y)$ are monotone under post-processing.*

Theorem S.8. *Let E be any symmetric encryption scheme, and let E' be the scheme obtained by uniformizing $p_{\mathrm{c|m,k}}$ on each equivalence class S of R_{a}. Then for any measure M that is monotone under post-processing, we have*

$$M(M,K;C') \leq M(M,K;C),\ \ M(M;C') \leq M(M;C),\ \ M(K;C') \leq M(K;C).$$

In particular,

$$I(K;C') \leq I(K;C), \quad \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{E}',p_{\mathrm{m}},p_{\mathrm{k}}} \leq \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{E},p_{\mathrm{m}},p_{\mathrm{k}}}, \quad \mathbf{Adv}^{\mathrm{tdss}}_{\mathsf{E}',p_{\mathrm{m}},p_{\mathrm{k}}} \leq \mathbf{Adv}^{\mathrm{tdss}}_{\mathsf{E},p_{\mathrm{m}},p_{\mathrm{k}}}.$$

Rearrangement of $p_{\mathrm{c|m,k}}$. After uniformizing $p_{\mathrm{c|m,k}}$ on $S_{\boldsymbol{m}}$, we aggregate the ciphertexts in $S_{\boldsymbol{m}}$ into a continuous interval on $\{0,1\}^l$. We then rearrange these intervals $S_{\boldsymbol{m}} = \cap_j \mathcal{C}^{(l)}(m_{i_j}, k_j)$ on $\mathcal{C}^{(l)}$ in lexicographic order of $(i_j)_j$, where $\boldsymbol{m} = (m_{i_j})_j$. As shown in Fig. S.6, the intervals are ordered as follows:

$$S_{(m_1,m_1,\ldots,m_1,m_1)}, S_{(m_1,m_1,\ldots,m_1,m_2)}, S_{(m_1,m_1,\ldots,m_1,m_3)}, \ldots,$$

$$S_{(m_1,m_1,\dots,m_2,m_1)}, S_{(m_1,m_1,\dots,m_2,m_2)}, S_{(m_1,m_1,\dots,m_2,m_3)}, \dots,$$
$$\dots,$$
$$S_{(m_2,m_1,\dots,m_1,m_1)}, S_{(m_2,m_1,\dots,m_1,m_2)}, S_{(m_2,m_1,\dots,m_1,m_3)}, \dots,$$
$$\dots.$$

Convergence of interval and $p_{\mathrm{c|m,k}}$. For a sequence of intervals $[a^{(l)}, b^{(l)})$ on $[0,1)$, both $a^{(l)}$ and $b^{(l)}$ must have convergent subsequences. Assume $a^{(l)}$ converges to a and $b^{(l)}$ converges to b. If $a \neq b$, then $p_{\mathrm{c|m,k}}$ naturally converges. However, if $a = b$, then $p_{\mathrm{c|m,k}}$ converges to an infinite probability density at a single point, which yields a degenerate asymptotic structure.

Fortunately, we can rearrange these intervals so that they converge to continuous intervals with finite probability density. Specifically, for intervals where $p_{\mathrm{c|m,k}} \to \infty$, $p_{\mathrm{c|m,k}}$ may exhibit different orders of infinity, such as $O(l)$ and $O(l^2)$. Due to the finite nature of the intervals, these orders are also finite. We label the order with the index i. For intervals $[a^{(l)}, b^{(l)})$ with the i-th order, we move them into $[i, i+1)$ and extend their lengths by the same factor, ensuring that $p_{\mathrm{c|m,k}}$ converges to a finite value.

Thus, the asymptotic scheme maintains finite probability density on a continuous ciphertext space $[0,n)$, which can then be naturally rescaled to $[0,1)$, where n is the number of orders.

We assert that this rearrangement does not affect the necessary conditions of asymptotic schemes, as the intersection of two intervals with different orders has a probability that converges to zero.

Corollary 1 (restated). *For a continuous encryption scheme E_{c} with $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{E}_{\mathrm{c}},p_{\mathrm{m}},p_{\mathrm{k}}} = p_{\max}$, the following conditions hold: Let $m_1 \in \arg\max_m p_{\mathrm{m}}(m)$ and $k_1 \in \arg\max_k p_{\mathrm{k}}(k)$.*

1. *In Case K with $p_{\mathrm{k}}^{\max} \geq p_{\mathrm{m}}^{\max}$: 1) $\{\mathcal{C}(m,k_1)\}_m$ is a partition of $\mathcal{C}$ and 2) $\mathcal{C}(m,k_1) \cap \mathcal{C}(m,k') = \emptyset$ for each m and each $k' \neq k_1$.*
2. *In Case M with $p_{\mathrm{m}}^{\max} \geq p_{\mathrm{k}}^{\max}$: $\cup_k \mathcal{C}(m_1,k) = \mathcal{C}$.*

Proof of Corollary 1. In Case K: if the condition does not hold, then there exists a ciphertext c such that 1) $D_{k_1}(c) = \perp$ or 2) $D_{k_1}(c) = D_{k'}(c)$ for some $k' \neq k_1$. For R_{a}, let S be the equivalence class including c, and $|S|$ be the interval length of S. If $D_{k_1}(c) = \perp$ holds, then the MRS advantage is not less than

$$p_{\mathrm{k}}(k_1) + |S| \max_m p_{\mathrm{m,c}}(m,c).$$

If $D_{k_1}(c) = D_{k'}(c) \in \mathcal{M}$ for some $k' \neq k_1$, then the MRS advantage is not less than

$$p_{\mathrm{k}}(k_1) + |S| p_{\mathrm{m,k,c}}(m_{\mathrm{triv}}(c), k', c).$$

In Case M, if the condition does not hold, then there exists a ciphertext c such that $D_k(c) \neq m_1$ for each k. Therefore, the MRS advantage is not less than

$$p_{\mathrm{m}}(m_1) + |S| \max_m p_{\mathrm{m,c}}(m,c).$$

Thus, the theorem holds. □

Theorem 12 (restated). *For a message distribution p_{m}, there exists a p_{k}-agnostic continuous encryption scheme E_{c} such that $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{E}_{\mathrm{c}},p_{\mathrm{m}},p_{\mathrm{k}}} = p_{\max}$ for any key distribution p_{k},* only *in the following three cases:*

1. *$|\mathcal{M}| = |\mathcal{K}| = 2$.*
2. *$|\mathcal{M}| \geq 3$, $|\mathcal{K}| = 2$, and $p_{\mathrm{m}}(m_1) \geq 0.5$.*
3. *$p_{\mathrm{m}}^{\max} \leq \frac{1}{|\mathcal{K}|}$.*

In other cases, such a p_{k}-agnostic scheme does not exist.

Proof of Theorem 12. If $\mathcal{M} = \mathcal{K} = \{0,1\}$, then $E_k(m) = m \oplus k$ with $\mathcal{C} = \{0,1\}$ achieves the optimal MRS. This scheme can be naturally extended to $[0,1)$, yielding E_2 (as shown in Fig. 8a).

If $|\mathcal{M}| \geq 3$, $|\mathcal{K}| = 2$, and $p_{\mathrm{m}}(m_1) \geq 0.5$, we use the scheme E_2 for $|\mathcal{M}| = 2$ and assign m_i for $i \neq 1$ within the intervals of m_2 in E_2 according to the probabilities (as shown in Fig. 8b). In other words, $\mathcal{C}(m_1,k) = \cup_{i\neq 1} \mathcal{C}(m_i,k')$ for each $k \neq k'$.

If $|\mathcal{M}| \geq 3$, $|\mathcal{K}| = 2$, and $p_{\mathrm{m}}(m_1) < 0.5$, then $p_{\mathrm{m}}(m_1) < \frac{1}{|\mathcal{K}|}$, which can be reduced to the next case.

If $p_{\mathrm{m}}(m_1) \leq \frac{1}{|\mathcal{K}|}$, we can use an EE construction with the following interval structure as the general scheme (as shown in Fig. 8c):

$$\mathcal{C}(m_i,k_j) = [F_{\mathrm{m}}(i-1) + (j-1)F_{\mathrm{m}}(1), F_{\mathrm{m}}(i) + (j-1)F_{\mathrm{m}}(1)).$$

Since $|\mathcal{K}| p_{\mathrm{m}}(m_1) \leq 1$, the scheme is well-defined and avoids collisions.

We now analyze the remaining case, namely $|\mathcal{K}| \geq 3$ and $p_{\mathrm{m}}(m_1) > \frac{1}{|\mathcal{K}|}$. In this case, general schemes do not exist. Otherwise, we derive contradictions by exploiting the arbitrariness of p_{k} and applying Corollary 1.

Let $p_{\mathrm{k}}(k')$ approach 1 for some k'. This corresponds to Case K, where $p_{\mathrm{k}}(k') > p_{\mathrm{m}}(m_1)$. Therefore, the following holds: 1) $\{\mathcal{C}(m,k')\}_m$ is a partition of $\mathcal{C}$, and 2) $\mathcal{C}(m,k') \cap \mathcal{C}(m,k'') = \emptyset$ for each $k'' \neq k'$ and each m. Since k' is arbitrary, each c must correspond to different messages under different keys. We next analyze two cases: 1) $p_{\mathrm{m}}(m_1) > \frac{1}{|\mathcal{K}|-1}$, and 2) $\frac{1}{|\mathcal{K}|} < p_{\mathrm{m}}(m_1) \leq \frac{1}{|\mathcal{K}|-1}$.

Case 1: $p_{\mathrm{m}}(m_1) > \frac{1}{|\mathcal{K}|-1}$:

Let $p_{\mathrm{k}}(k)$ be uniform. Since $p_{\mathrm{m}}(m_1) > \frac{1}{|\mathcal{K}|}$, this corresponds to Case M. Therefore, $\{\mathcal{C}(m_1,k)\}_k$ covers $\mathcal{C}$. Furthermore, by the above analysis that each c corresponds to different m under different k, $\{\mathcal{C}(m_1,k)\}_k$ forms a partition of $\mathcal{C}$. Then, let p_{k} be uniform over $\mathcal{K}\backslash\{k'\}$ for some k' and $p_{\mathrm{k}}(k') = 0$. As $|\mathcal{K}| \geq 3$, this yields a general scheme on $\mathcal{K}\backslash\{k'\}$. Because $p_{\mathrm{m}}(m_1) > \frac{1}{|\mathcal{K}|-1}$, it still corresponds to Case M. However, $\{\mathcal{C}(m_1,k)\}_{k\neq k'}$ no longer forms a partition of $\mathcal{C}$, which contradicts Corollary 1. Thus, general schemes do not exist.

Case 2: $\frac{1}{|\mathcal{K}|} < p_{\mathrm{m}}(m_1) \leq \frac{1}{|\mathcal{K}|-1}$: Let $p_{\mathrm{k}}(k_i) = \frac{1}{2} = p_{\mathrm{k}}(k_j)$ for some $k_i \neq k_j$. Since $p(m_1) \leq \frac{1}{|\mathcal{K}|-1} \leq \frac{1}{3-1} = \frac{1}{2}$, this falls into Case K. By Theorem 11, $p_{\mathrm{m,c}}(D_{k_i}(c),c) = p_{\mathrm{m,c}}(D_{k_j}(c),c)$ for each c.

Given that each c corresponds to a different m under different k, it holds that $p_{\mathrm{m,c}}(D_k(c),c) = p_{\mathrm{k}}(k)p_{\mathrm{m}}(D_k(c))p_{\mathrm{c|m,k}}(c \,|\, D_k(c),k)$. Thus, due to the arbitrariness of k_i and k_j, the product $p_{\mathrm{m}}(D_k(c)) \cdot p_{\mathrm{c|m,k}}(c \,|\, D_k(c),k)$ must be constant for all k and all c in any equivalence class S of R_{a}. It is easy to stretch equivalence classes such that this product remains constant for all k and c.

Moreover,

$$\sum_c p_{\mathrm{m}}(D_k(c))p_{\mathrm{c|m,k}}(c \,|\, D_k(c),k) = \sum_m p_{\mathrm{m}}(m) \sum_{c\in\mathcal{C}(m,k)} p_{\mathrm{c|m,k}}(c \,|\, m,k) = 1,$$

implying $p_{\mathrm{m}}(D_k(c))p_{\mathrm{c|m,k}}(c \,|\, D_k(c),k) = 1$.

Now, let $p_{\mathrm{k}}(k) = \frac{1}{|\mathcal{K}|}$ for each k. Since $p_{\mathrm{m}}(m_1) > \frac{1}{|\mathcal{K}|}$, this corresponds to Case M. By Corollary 1 and the previously established fact that $\mathcal{C}(m,k') \cap \mathcal{C}(m,k'') = \emptyset$ for all $k'' \neq k'$ and each m, the family $\{\mathcal{C}(m_1,k)\}_k$ forms a partition of $\mathcal{C}$.

Hence, for each c, there exists a unique k such that $D_k(c) = m_1$. Denote this k as $k(c,m_1)$. Then,

$$p_{\mathrm{m}}(m_1)p_{\mathrm{c|m,k}}(c \,|\, m_1, k(c,m_1)) = 1,$$

which gives

$$p_{\mathrm{c|m,k}}(c \,|\, m_1, k(c,m_1)) = \frac{1}{p_{\mathrm{m}}(m_1)} = \frac{1}{|\mathcal{C}(m_1,k(c,m_1))|}.$$

Thus, $|\mathcal{C}(m_1,k(c,m_1))| \cdot |\mathcal{K}| > 1$, contradicting that $\{\mathcal{C}(m_1,k)\}_k$ is a partition of $\mathcal{C}$. □

Proposition 8 (restated). *For a message distribution p_{m}, there exists a continuous scheme E_{c} such that $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{E}_{\mathrm{c}},p_{\mathrm{m}},p_{\mathrm{k}}} = p_{\max}$ for any key distribution p_{k} with $p_{\mathrm{k}}^{\max} \leq \frac{1}{|\mathcal{M}|}$.*

Proof of Proposition 8. This proposition corresponds only to Case M.

In that case, HES for uniform message distribution suffices. □

Lemma S.1. *For $f_a(x) = \max\{\frac{a}{x}, \frac{1-a}{1-x}\} + \max\{\frac{a}{1-x}, \frac{1-a}{x}\}$, it always holds*

$$f_{a_1}(x) > f_{a_2}(x)$$

for $0 < a_1 < a_2 < 1/2$ and $0 < x < 1$.

Example 1 (restated). *For $p_{\mathrm{m}} = (\frac{1}{2}, \frac{1}{2})$ and $p_{\mathrm{k}} = (p_1, p_2, p_3)$ with $\frac{1}{2} > p_1 \geq p_2 \geq p_3$:*

1. *If $p_2 \geq \frac{1}{3}$,*

$$\inf_{\mathsf{EE}\in\Omega_{EE}} \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}} = \min\left\{\frac{1-2p_1}{2}, \frac{(2p_1-1)(1-3p_1)}{2(1-p_1)p_1}, \frac{1-3p_3}{4}, \frac{(1-2p_2)(p_2-p_3)}{2(1-p_2)p_2}\right\} + \frac{1}{2}.$$

2. *If $p_2 \leq \frac{1}{3}$,*

$$\inf_{\mathsf{EE}\in\Omega_{EE}} \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}} = \min\left\{\tfrac{1-2p_1}{2},\ \tfrac{3p_1-1}{4}\right\} + \tfrac{1}{2}.$$

Then, $\inf_{\mathsf{EE}\in\Omega_{EE}} \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}} \geq \frac{1}{2}$, where the equality holds if and only if $p_1 = p_2 = p_3 = \frac{1}{3}$.

Derivation of Example 1. In an EE construction, for each key k the ciphertext distribution is uniform over $\mathcal{C}$, and for every message m the size of the set $\mathcal{C}(m,k)$ is independent of k. Consequently, EE constructions can freely adjust the sizes of the equivalence classes

$$S_{(m_{i_1},m_{i_2},m_{i_3})}$$

while preserving the uniformity and constant size of each $\mathcal{C}(m_i, k)$.

For notational convenience, write

$$s_{i_1 i_2 i_3} \;=\; \big|S_{(m_{i_1}, m_{i_2}, m_{i_3})}\big|, \qquad \alpha_i \;=\; \big|\mathcal{C}(m_i, k)\big|.$$

These variables satisfy

$$\begin{aligned}
s_{111} + s_{112} + s_{121} + s_{122} &= \alpha_1, \\
s_{111} + s_{112} + s_{211} + s_{212} &= \alpha_1, \\
s_{111} + s_{121} + s_{211} + s_{221} &= \alpha_1, \\
\sum_{i_1, i_2, i_3 = 1}^{2} s_{i_1 i_2 i_3} &= 1, \\
\alpha_1 + \alpha_2 &= 1, \\
s_{i_1 i_2 i_3} &\geq 0 \quad \forall\, i_1, i_2, i_3 \in \{1, 2\}.
\end{aligned}$$

First, observe that setting

$$s_{111} = s_{222} = 0$$

cannot increase the adversary's advantage. Indeed, any positive mass on s_{111} or s_{222} contributes an advantage of 1, and one can redistribute this mass to other classes to lower the overall advantage.

Hence, without loss of generality we set $s_{111} = s_{222} = 0$ and treat α_1, s_{112}, and s_{121} as free parameters. The remaining variables admit the closed-form expressions

$$\begin{aligned}
s_{122} &= \alpha_1 - s_{112} - s_{121}, \\
s_{211} &= -1 + 3\,\alpha_1 - s_{112} - s_{121}, \\
s_{212} &= 1 - 2\,\alpha_1 + s_{121}, \\
s_{221} &= 1 - 2\,\alpha_1 + s_{112}, \\
\alpha_2 &= 1 - \alpha_1,
\end{aligned}$$

subject to

$$\frac{1}{3} \;\leq\; \alpha_1 \;\leq\; \frac{2}{3}, \qquad L \;\leq\; s_{112} \;\leq\; U, \qquad L \;\leq\; s_{121} \;\leq\; U,$$

where

$$L = \max\{0,\; 2\alpha_1 - 1\}, \qquad U = \min\{\alpha_1,\; 3\alpha_1 - 1\}.$$

For any $c \in S_{i_1 i_2 i_3}$,

$$\max_m \Pr[M = m,\, C = c] = \frac{1}{2}\; \max\left\{ \frac{\sum_{j: i_j = 1} p_j}{\alpha_1},\; \frac{\sum_{j: i_j = 2} p_j}{\alpha_2} \right\}.$$

Therefore the overall advantage is

$$\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE}, p_{\mathrm{m}}, p_{\mathrm{k}}} = \sum_{i_1, i_2, i_3} \frac{s_{i_1 i_2 i_3}}{2}\; \max\left\{ \frac{\sum_{j: i_j = 1} p_j}{\alpha_1},\; \frac{\sum_{j: i_j = 2} p_j}{\alpha_2} \right\}.$$

Fix α_1, the feasible region for $(s_{i_1 i_2 i_3})_{i_1 i_2 i_3}$ is a triangle, and $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE}, p_{\mathrm{m}}, p_{\mathrm{k}}}$ is affine in these two variables as

$$\frac{1}{2}\, s_{121}\, A_1 \;+\; \frac{1}{2}\, s_{112}\, A_2 \;+\; \frac{1}{2}\, A_3,$$

where

$$\begin{aligned}
A_1 = &\max\left\{ \frac{p_2}{\alpha_1},\, \frac{1 - p_2}{1 - \alpha_1} \right\} + \max\left\{ \frac{p_2}{1 - \alpha_1},\, \frac{1 - p_2}{\alpha_1} \right\} \\
&- \max\left\{ \frac{p_1}{\alpha_1},\, \frac{1 - p_1}{1 - \alpha_1} \right\} - \max\left\{ \frac{p_1}{1 - \alpha_1},\, \frac{1 - p_1}{\alpha_1} \right\}, \\
A_2 = &\max\left\{ \frac{p_3}{\alpha_1},\, \frac{1 - p_3}{1 - \alpha_1} \right\} + \max\left\{ \frac{1 - p_3}{\alpha_1},\, \frac{p_3}{1 - \alpha_1} \right\}
\end{aligned}$$

$$
\begin{aligned}
&- \max\left\{\frac{p_1}{\alpha_1},\, \frac{1-p_1}{1-\alpha_1}\right\} - \max\left\{\frac{p_1}{1-\alpha_1},\, \frac{1-p_1}{\alpha_1}\right\},\\
A_3 =& \alpha_1 \max\left\{\frac{p_1}{\alpha_1},\, \frac{1-p_1}{1-\alpha_1}\right\} + \max\left\{\frac{p_2}{\alpha_1},\, \frac{1-p_2}{1-\alpha_1}\right\} + \max\left\{\frac{p_3}{\alpha_1},\, \frac{1-p_3}{1-\alpha_1}\right\}\\
&+ 3\,\alpha_1 \max\left\{\frac{p_1}{1-\alpha_1},\, \frac{1-p_1}{\alpha_1}\right\} - 2\,\alpha_1 \max\left\{\frac{p_2}{\alpha_1},\, \frac{1-p_2}{1-\alpha_1}\right\}\\
&- 2\,\alpha_1 \max\left\{\frac{p_3}{\alpha_1},\, \frac{1-p_3}{1-\alpha_1}\right\} - \max\left\{\frac{p_1}{1-\alpha_1},\, \frac{1-p_1}{\alpha_1}\right\}.
\end{aligned}
$$

As $1 > p_1 \geq p_2 \geq p_3 \geq 0$, by Lemma S.1, $A_1 \geq 0$ and $A_2 \geq 0$. Therefore, $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}}$ achieves its minimum value A_3 when

$$s_{112} = s_{121} = L.$$

At this vertex $s_{i_1 i_2 i_3}$ take the values

$$
\begin{aligned}
s_{111} &= s_{222} = 0,\\
s_{122} &= \alpha_1 - 2L,\\
s_{211} &= 3\,\alpha_1 - 1 - 2L,\\
s_{212} &= 1 - 2\,\alpha_1 + L,\\
s_{221} &= 1 - 2\,\alpha_1 + L.
\end{aligned}
$$

Naturally, we can partition the space $\left[\frac{1}{3}, \frac{2}{3}\right]$ of α_1 into several intervals such that, within each interval, each expression of the form $\max\{\cdot,\cdot\}$ becomes fixed. Then,

$$\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}} = A + \frac{B}{\alpha_1} + \frac{C}{1-\alpha_1},$$

where A, B, C are constants determined by (p_1, p_2, p_3). *We can compute the minimum of $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}}$ over each interval and then determine the global minimum.*

The interval boundaries consist of all switching points of the $\max$ functions, along with the endpoints of the interval $\left[\frac{1}{3}, \frac{2}{3}\right]$, forming the set

$$\left\{\frac{1}{3},\, \frac{1}{2},\, \frac{2}{3},\, p_1,\, p_2,\, p_3,\, p_1{+}p_2,\, p_1{+}p_3,\, p_2{+}p_3\right\} \cap \left[\frac{1}{3}, \frac{2}{3}\right].$$

Considering the symmetry between m_1 and m_2, we can construct an EE' with $s'_{i'_1 i'_2 i'_3} = s_{i_1 i_2 i_3}$, $i'_j = 3{-}i_j$ and $\alpha'_i = 1{-}\alpha$ such that

$$\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE}',p_{\mathrm{m}},p_{\mathrm{k}}} = \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}}.$$

Thus, $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}}$ achieves the same value at α_1 and $1-\alpha_1$. We only need to consider the minimum on the interval $\left[\frac{1}{3}, \frac{1}{2}\right]$ with the vertex satisfying

$$
\begin{aligned}
s_{111} &= s_{222} = s_{112} = s_{121} = 0,\\
s_{122} &= \alpha_1,\\
s_{211} &= 3\,\alpha_1 - 1,\\
s_{212} &= s_{221} = 1 - 2\,\alpha_1,
\end{aligned}
$$

and the boundary point set

$$P = \left\{\frac{1}{3},\, \frac{1}{2},\, p_1,\, p_2,\, p_3\right\} \cap \left[\frac{1}{3}, \frac{1}{2}\right].$$

In fact, for $0 < \alpha_1 < 1$, it always holds that $B \leq 0$ and $C \leq 0$ (see the following calculation). Therefore, $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}}$ is a concave function of α_1 and its minimum occurs at one of the interval endpoints.

P varies in two cases: 1) $p_2 \geq \frac{1}{3}$ and 2) $p_2 \leq \frac{1}{3}$.

Case 1: If $p_2 \geq \frac{1}{3}$, then

$$p_3 = 1 - p_1 - p_2 \leq \frac{1}{3} \leq p_2 \leq p_1 < \frac{1}{2}.$$

Thus,

$$P = \left\{\frac{1}{3},\, p_2,\, p_1,\, \frac{1}{2}\right\}.$$

On each resulting interval, $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}}$ simplifies to one of the following expressions:

$$\frac{3}{2}-\frac{1-p_3}{2(1-\alpha_1)}-\frac{p_3}{2\alpha_1},$$
$$\frac{5}{2}-\frac{1-p_1}{2\alpha_1}-\frac{p_1+1}{2(1-\alpha_1)},$$
$$2-\frac{1-p_1}{2\alpha_1}-\frac{p_1}{1-\alpha_1}.$$

Thus, we compute the values on the endpoints:

$$\frac{1}{4}(1-3p_3)+\frac{1}{2},$$
$$\frac{(2p_2-1)(p_1+2p_2-1)}{2(p_2-1)p_2}+\frac{1}{2},$$
$$\frac{(2p_1-1)(3p_1-1)}{2(p_1-1)p_1}+\frac{1}{2},$$
$$\frac{1}{2}(1-2p_1)+\frac{1}{2}.$$

Finally, the global minimum is given by:

$$\min\left\{\frac{1-2p_1}{2},\ \frac{(2p_1-1)(1-3p_1)}{2(1-p_1)p_1},\ \frac{1-3p_3}{4},\ \frac{(1-2p_2)(p_2-p_3)}{2(1-p_2)p_2}\right\}+\frac{1}{2}.$$

Case 2: If $p_2 \leq \frac{1}{3}$, then

$$p_3 \leq p_2 \leq \frac{1}{3} \leq p_1 \leq \frac{1}{2}.$$

Therefore,

$$P=\left\{\frac{1}{3},\ p_1,\ \frac{1}{2}\right\}.$$

On the resulting intervals, the expression for $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}}$ becomes:

$$\frac{5}{2}-\frac{1-p_1}{2\alpha_1}-\frac{p_1+1}{2(1-\alpha_1)},$$
$$2-\frac{1-p_1}{2\alpha_1}-\frac{p_1}{1-\alpha_1}.$$

We evaluate the expression at the endpoints:

$$\frac{3p_1-1}{4}+\frac{1}{2},$$
$$\frac{(2p_1-1)(3p_1-1)}{2(p_1-1)p_1}+\frac{1}{2},$$
$$\frac{1-2p_1}{2}+\frac{1}{2}.$$

Thus, the minimum value of $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE},p_{\mathrm{m}},p_{\mathrm{k}}}$ is:

$$\min\left\{\frac{1-2p_1}{2},\ \frac{3p_1-1}{4}\right\}+\frac{1}{2}.$$

□

Example 2 (restated). *For $p_{\mathrm{m}}=(\frac{1}{2},\frac{1}{2})$ and $p_{\mathrm{k}}=(p_1,p_2,p_3)$ with $\frac{1}{2}>p_1\geq p_2\geq p_3$:*

$$\inf_{\mathsf{HE}\in\Omega_{HE}}\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{HE},p_{\mathrm{m}},p_{\mathrm{k}}}=\tfrac{1-2p_1}{2}+\tfrac{1}{2}>\tfrac{1}{2}.$$

Derivation of Example 2. By the derivation of Example 1, setting $\alpha_1=\frac{1}{2}$ naturally implies this bound. □

Theorem 16 (restated). *For any message and key distributions* $(p_{\mathrm{m}}, p_{\mathrm{k}})$, HES *satisfies*

$$I(K;C) = 0 \quad \textit{and} \quad \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{HES},p_{\mathrm{m}},p_{\mathrm{k}}} \leq p_{\mathrm{k}}^{\max} + \tfrac{1-p_{\mathrm{k}}^{\max}}{\lfloor 1/p_{\mathrm{m}}^{\max} \rfloor} \mathbf{1}_{1/p_{\mathrm{m}}^{\max} < |\mathcal{K}|} \leq 2p_{\max}.$$

Proof of Theorem 16. Without loss of generality, assume $p_{\max} < \frac{1}{2}$ and $1/p_{\mathrm{m}}^{\max} < |\mathcal{K}|$, since otherwise the claim is trivial (see the proof of Theorem 12).

Let $t = \lfloor 1/p_{\mathrm{m}}(m_1) \rfloor$ and partition the key space $\mathcal{K}$ into $\lceil N_{\mathrm{k}}/t \rceil$ groups, where the i-th group is $\mathcal{K}_i = \{k_{ti+1}, \ldots, k_{ti+t}\}$ (the last group ends with $k_{N_{\mathrm{k}}}$).

By construction of HES, there are no collisions within any group $\mathcal{K}_i$: for any $k_1, k_2 \in \mathcal{K}_i$ and $c \in \mathcal{C}$, we have $D_{k_1}(c) \neq D_{k_2}(c)$. Since HES is a HE scheme, $p_{\mathrm{c}|\mathrm{m},\mathrm{k}}(c|m,k) = 1/p_{\mathrm{m}}(m)$. Hence,

$$\begin{aligned} p_{\mathrm{m},\mathrm{c}}(m,c) &= \sum_{k:D_k(c)=m} p_{\mathrm{k}}(k) p_{\mathrm{m}}(m) p_{\mathrm{c}|\mathrm{m},\mathrm{k}}(c|m,k) \\ &= \sum_{k:D_k(c)=m} p_{\mathrm{k}}(k) \leq \sum_{i=0}^{\lceil N_{\mathrm{k}}/t \rceil - 1} p_{\mathrm{k}}(k_{ti+1}). \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{HES},p_{\mathrm{m}},p_{\mathrm{k}}} &\leq \sum_{i=0}^{\lceil N_{\mathrm{k}}/t \rceil - 1} p_{\mathrm{k}}(k_{ti+1}) \\ &\leq p_{\mathrm{k}}(k_1) + \sum_{i=1}^{\lceil N_{\mathrm{k}}/t \rceil - 1} \tfrac{1}{t} \sum_{j=0}^{t-1} p_{\mathrm{k}}(k_{ti+1-j}) \\ &\leq p_{\mathrm{k}}(k_1) + \tfrac{1}{t} \sum_{i=2}^{N_{\mathrm{k}}} p_{\mathrm{k}}(k_i) \\ &= p_{\mathrm{k}}(k_1) + \tfrac{1-p_{\mathrm{k}}(k_1)}{t}. \end{aligned}$$

Since $t = \lfloor 1/p_{\mathrm{m}}(m_1) \rfloor \geq 1/p_{\mathrm{m}}(m_1) - 1$, we get

$$\begin{aligned} p_{\mathrm{k}}(k_1) + \tfrac{1-p_{\mathrm{k}}(k_1)}{t} &\leq p_{\mathrm{k}}(k_1) + \frac{1-p_{\mathrm{k}}(k_1)}{1/p_{\mathrm{m}}(m_1) - 1} \\ &= \frac{p_{\mathrm{k}}(k_1) + p_{\mathrm{m}}(m_1) - 2p_{\mathrm{k}}(k_1)p_{\mathrm{m}}(m_1)}{1-p_{\mathrm{m}}(m_1)} \\ &\leq \frac{p_{\max} + (1-2p_{\max})p_{\min}}{1-p_{\max}} \\ &\leq \frac{p_{\max} + (1-2p_{\max})p_{\max}}{1-p_{\max}} = 2p_{\max}, \end{aligned}$$

where $p_{\max} = \max\{p_{\mathrm{k}}(k_1), p_{\mathrm{m}}(m_1)\}$ and $p_{\min} = \min\{p_{\mathrm{k}}(k_1), p_{\mathrm{m}}(m_1)\}$. □

Proposition 10 (restated). *For any EE construction* $\mathsf{EE}^{(l)}$ *in the random oracle or ideal cipher models and the associated continuous scheme* EE_{c}, *as* $l \to \infty$,

$$\mathbb{E}[I_{\mathsf{EE}^{(l)}}(K;C^{(l)})] \to I_{\mathsf{EE}_{\mathrm{c}}}(K;C) \quad \textit{and} \quad \mathbb{E}[\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE}^{(l)},p_{\mathrm{m}},p_{\mathrm{k}}}] \to \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE}_{\mathrm{c}},p_{\mathrm{m}},p_{\mathrm{k}}}.$$

Proof of Proposition 10. In an EE construction, $p_{\mathrm{c}|\mathrm{m},\mathrm{k}}(c|m,k) = \frac{1}{2^l p_{\mathrm{d}}^{(l)}(m)}$, so $p_{\mathrm{c}|\mathrm{k}}(c|k)$, $p_{\mathrm{c}}(c)$, and $p_{\mathrm{m},\mathrm{c}}(m,c)$ are constant for all $c \in S_{\boldsymbol{m}}$. Therefore,

$$\begin{aligned} I(K;C) &= \sum_k p_{\mathrm{k}}(k) \sum_c p_{\mathrm{c}|\mathrm{k}}(c|k) \log \frac{p_{\mathrm{c}|\mathrm{k}}(c|k)}{p_{\mathrm{c}}(c)} \\ &= \sum_{\boldsymbol{m}} |S_{\boldsymbol{m}}| \sum_k p_{\mathrm{k}}(k) p_{\mathrm{c}|\mathrm{k}}(c|k) \log \frac{p_{\mathrm{c}|\mathrm{k}}(c|k)}{p_{\mathrm{c}}(c)}. \end{aligned}$$

$$\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE}^{(l)},p_{\mathrm{m}},p_{\mathrm{k}}} = \sum_c \max_m p_{\mathrm{m},\mathrm{c}}(m,c) = \sum_{\boldsymbol{m}} |S_{\boldsymbol{m}}| \max_m p_{\mathrm{m},\mathrm{c}}(m,c).$$

It suffices to compute $\mathbb{E}[|S_{\boldsymbol{m}}|]$ in the RO/IC setting. Let I_c be the indicator that $D_{k_i}(c) = m_{j_i}$ for all i. Then

$$\mathbb{E}[I_c] = \Pr[D_{k_i}(c) = m_{j_i}, \forall i] = \Pr[\mathrm{decode}(c \oplus H(k_i)) = m_{j_i}, \forall i].$$

Since $c \oplus H(k_i)$ is independently uniform over $\{0,1\}^l$ in the RO/IC models,

$$\mathbb{E}[|S_{\boldsymbol{m}}|] = \sum_c \mathbb{E}[I_c] = \sum_c \Pr[\mathrm{decode}(c \oplus H(k_i)) = m_{j_i}, \forall i] = 2^l \prod_i p_{\mathrm{d}}^{(l)}(m_{j_i}).$$

As $p_{\mathrm{d}}^{(l)}(m_{j_i}) \to p_{\mathrm{d}}(m_{j_i})$, we set $|S_{\boldsymbol{m}}| = \prod_i p_{\mathrm{d}}(m_{j_i})$ and $p_{\mathrm{c|m,k}}(c \mid m, k) = 1/p_{\mathrm{d}}(m)$ in the continuous scheme EE_{c}, which yields the claimed convergence. □

Example 3 (restated). *For $p_{\mathrm{m}} = \left(\frac{1}{2}, \frac{1}{2}\right)$ and $p_{\mathrm{k}} = (p_1, p_2, p_3)$ with $\frac{1}{2} > p_1 \geq p_2 \geq p_3$, in the random-oracle or ideal-cipher model, the continuous HE scheme HE_{c} (with $p_{\mathrm{d}} = p_{\mathrm{m}}$) and EE constructions EE_{c} (with general p_{d}) satisfy the bounds:*

$$\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{HE}_{\mathrm{c}}, p_{\mathrm{m}}, p_{\mathrm{k}}} = \frac{3}{4} \quad \textit{and} \quad \inf_{\mathsf{EE}_{\mathrm{c}} \in \Omega_{EE}} \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE}_{\mathrm{c}}, p_{\mathrm{m}}, p_{\mathrm{k}}} = \frac{1}{2} + p_1(1 - p_1) \geq \frac{13}{18}.$$

Derivation of Example 3. We borrow the notation from the derivation of Example 1. By Proposition 10, we have $s_{i_1 i_2 i_3} = \alpha_{i_1} \alpha_{i_2} \alpha_{i_3}$ where $i_1, i_2, i_3 \in \{1, 2\}$.

For the HE scheme, we have $\alpha_1 = \alpha_2 = \frac{1}{2}$, so $s_{i_1 i_2 i_3} = \frac{1}{8}$, and:

$$\begin{aligned}
&\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{HE}_{\mathrm{c}}, p_{\mathrm{m}}, p_{\mathrm{k}}} \\
&= \sum_{i_1, i_2, i_3} \frac{s_{i_1 i_2 i_3}}{2} \max \left\{ \frac{\sum_{j: i_j = 1} p_j}{\alpha_1}, \frac{\sum_{j: i_j = 2} p_j}{\alpha_2} \right\} \\
&= \frac{1}{8} \Big(\max\{1, 0\} + \max\{1 - p_3, p_3\} + \max\{1 - p_2, p_2\} + \max\{1 - p_1, p_1\} \\
&\quad + \max\{p_1, 1 - p_1\} + \max\{p_2, 1 - p_2\} + \max\{p_3, 1 - p_3\} + \max\{0, 1\} \Big) \\
&= \frac{1}{8} \left(1 + 1 - p_3 + 1 - p_2 + 1 - p_1 + 1 - p_1 + 1 - p_2 + 1 - p_3 + 1\right) \\
&= \frac{3}{4}.
\end{aligned}$$

For EE constructions, suppose $\alpha_1 = \alpha, \alpha_2 = 1 - \alpha$ and $0 < \alpha \leq \frac{1}{2}$. We have:

$$\begin{aligned}
&\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE}_{\mathrm{c}}, p_{\mathrm{m}}, p_{\mathrm{k}}} \\
&= \sum_{i_1, i_2, i_3} \frac{s_{i_1 i_2 i_3}}{2} \max \left\{ \frac{\sum_{j: i_j = 1} p_j}{\alpha}, \frac{\sum_{j: i_j = 2} p_j}{1 - \alpha} \right\} \\
&= \frac{\alpha^3}{2} \max \left\{ \frac{1}{\alpha}, 0 \right\} \\
&\quad + \frac{\alpha^2(1 - \alpha)}{2} \left(\max \left\{ \frac{1 - p_3}{\alpha}, \frac{p_3}{1 - \alpha} \right\} + \max \left\{ \frac{1 - p_2}{\alpha}, \frac{p_2}{1 - \alpha} \right\} \right. \\
&\quad \left. + \max \left\{ \frac{1 - p_1}{\alpha}, \frac{p_1}{1 - \alpha} \right\} \right) \\
&\quad + \frac{\alpha(1 - \alpha)^2}{2} \left(\max \left\{ \frac{p_1}{\alpha}, \frac{1 - p_1}{1 - \alpha} \right\} + \max \left\{ \frac{p_2}{\alpha}, \frac{1 - p_2}{1 - \alpha} \right\} \right. \\
&\quad \left. + \max \left\{ \frac{p_3}{\alpha}, \frac{1 - p_3}{1 - \alpha} \right\} \right) + \frac{(1 - \alpha)^3}{2} \max \left\{ 0, \frac{1}{1 - \alpha} \right\} \\
&= \frac{\alpha^2}{2} + \frac{\alpha^2(1 - \alpha)}{2} \left(\frac{1 - p_3}{\alpha} + \frac{1 - p_2}{\alpha} + \frac{1 - p_1}{\alpha} \right) \\
&\quad + \frac{1 - \alpha}{2} \left(\max\{p_1, \alpha\} - p_1 \alpha + \max\{p_2, \alpha\} - p_2 \alpha + \max\{p_3, \alpha\} - p_3 \alpha \right) \\
&\quad + \frac{(1 - \alpha)^2}{2} \\
&= \frac{1}{2} + \frac{1 - \alpha}{2} \left(\max\{p_1, \alpha\} + \max\{p_2, \alpha\} + \max\{p_3, \alpha\} - \alpha \right)
\end{aligned}$$

$$= \begin{cases} \frac{\alpha^2}{2} - \alpha + 1, & 0 < \alpha \le p_3 \\ \frac{1+(1-\alpha)(1-p_3)}{2}, & p_3 < \alpha \le p_2 \\ \frac{1+(1-\alpha)(p_1+\alpha)}{2}, & p_2 < \alpha \le p_1 \\ \frac{1}{2} + \alpha(1-\alpha), & p_1 < \alpha \le \frac{1}{2} \end{cases}.$$

By simple analysis of the above piecewise function of α, the minimum value is achieved when $\alpha = p_1$, therefore:

$$\inf_{\mathsf{EE} \in \Omega_{\mathrm{EE}}} \mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{EE}_{\mathrm{c}}, p_{\mathrm{m}}, p_{\mathrm{k}}} = \frac{1}{2} + p_1(1-p_1).$$

Moreover, the above results show that the optimal encoder is $p_{\mathrm{d}} = (p_1, 1-p_1)$ *but not* $\left(\frac{1}{2}, \frac{1}{2}\right)$ *as in the HE scheme.* □

Theorem 15 (restated). *There exist distributions* p_{m} *and* p_{k} *such that no continuous scheme* E_{c} *can satisfy both* $I(K;C) = 0$ *and* $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{E}_{\mathrm{c}}, p_{\mathrm{m}}, p_{\mathrm{k}}} = p_{\max}$.

Proof of Theorem 15. Similar to the analysis of key confidentiality (KC) for OE in Section 5.6, we leverage the necessary conditions for KC and MRS to prove the theorem.

Let p_{m} be the uniform distribution over $\{m_1, m_2\}$, and let p_{k} be the uniform distribution over $\{k_1, k_2, k_3\}$. This setting corresponds to Case M, since $p_{\mathrm{m}}(m) = \frac{1}{2} > \frac{1}{3} = p_{\mathrm{k}}(k)$.

Assume that such an encryption scheme exists and denote it by E. By Theorem 9, we have $p_{\mathrm{c|k}}(c\,|\,k) = p_{\mathrm{c}}(c)$. Since E is piecewise uniform, it can be stretched to make p_{c} uniform over the entire ciphertext space.

By Theorem 11, both m_1 and m_2 must maximize

$$p_{\mathrm{m,c}}(m, c) = \sum_{k: D_k(c) = m} p_{\mathrm{m}}(m) \cdot p_{\mathrm{k}}(k) \cdot p_{\mathrm{c|k}}(c\,|\,k) = \frac{1}{6} \cdot |\{k\,|\,D_k(c) = m\}|$$

for each c. However, since $\sum_m |\{k\,|\,D_k(c) = m\}| = 3$, it is impossible for both $|\{k\,|\,D_k(c) = m_1\}|$ and $|\{k\,|\,D_k(c) = m_2\}|$ to equal $\frac{3}{2}$. Therefore, no such encryption scheme E exists, which completes the proof. □

S.10. Sufficiency and Necessity for Optimal Security

Based on the analysis in Sections 4 and 5, we summarize the necessary and sufficient conditions for achieving optimal security. By categorizing different spaces of encryption schemes, we clarify these conditions: sufficient conditions are determined by their feasibility within these spaces, while necessary conditions are identified by whether optimally secure schemes exist within these specific spaces.

For KC, let $I_{p_{\mathrm{m}}, p_{\mathrm{k}}}(\Omega)$ represent the infimum of $I_{\mathsf{E}}(C, K)$ across all encryption schemes $\mathsf{E} \in \Omega$ for the given p_{m} and p_{k}. For message confidentiality, denote $\mathbf{Adv}^{\mathrm{mrs}}_{\Omega, p_{\mathrm{m}}, p_{\mathrm{k}}}$ as the infimum of the advantage $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{E}, p_{\mathrm{m}}, p_{\mathrm{k}}}$ over all schemes $\mathsf{E} \in \Omega$. For a family E of encryption schemes, where each scheme is parameterized by l as $\mathsf{E}^{(l)} = (\mathcal{M}, \mathcal{K}, \mathcal{C}^{(l)}, \mathcal{E}^{(l)}, \mathcal{D}^{(l)})$, the infima of $I_{\mathsf{E}^{(l)}}(C, K)$ and $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{E}^{(l)}, p_{\mathrm{m}}, p_{\mathrm{k}}}$ are respectively denoted as $I_{\mathsf{E}}(C, K)$ and $\mathbf{Adv}^{\mathrm{mrs}}_{\mathsf{E}, p_{\mathrm{m}}, p_{\mathrm{k}}}$.

With the previous analysis in Sections 4 and 5, on the spaces Ω_{E} of symmetric encryption, for arbitrary message and key distributions $(p_{\mathrm{m}}, p_{\mathrm{k}})$, we have the optimal security with $I_{p_{\mathrm{m}}, p_{\mathrm{k}}}(\Omega_{\mathrm{E}}) = 0$ and $\mathbf{Adv}^{\mathrm{mrs}}_{\Omega, p_{\mathrm{m}}, p_{\mathrm{k}}} = p_{\max}$.

We consider the spaces of deterministic encryption, probabilistic encryption, EE constructions and HE schemes, denoted as Ω_{DE}, Ω_{PE}, Ω_{EE}, and Ω_{HE}, respectively.

Moreover, we analyze probabilistic encryption schemes characterized by different attributes: 1) ciphertext uniformity (U), 2) ciphertext cardinality dependence or independence from the key (KD or KID), and 3) ciphertext cardinality dependence or independence from the message (MD or MID). We denote these spaces as Ω^{P}_{PE} for attributes P, where P = U, KD, KID, MD, MID, or combinations thereof.

We also discuss the spaces of symmetric encryption schemes achieving optimal security for MC and KC, denoted as $\Omega^{\mathrm{MC}}_{\mathrm{E}}$ and $\Omega^{\mathrm{KC}}_{\mathrm{E}}$, respectively.

Delving into the insights from Section 4, we have:

1. For general non-uniform message distributions p_{m}, it typically holds that $I_{p_{\mathrm{m}}, p_{\mathrm{k}}}(\Omega_{\mathrm{DE}}) > 0$ and $I_{p_{\mathrm{m}}, p_{\mathrm{k}}}(\Omega^{\mathrm{U,MID}}_{\mathrm{PE}}) > 0$. In other words, it generally follows that $\Omega^{\mathrm{KC}}_{\mathrm{E}} \cap \Omega^{\mathrm{U}}_{\mathrm{PE}} \subseteq \Omega^{\mathrm{U,MD}}_{\mathrm{PE}}$. This implies that uniform probabilistic encryption with optimal KC must be message-dependent. Without the constraint of uniformity, this proposition does not hold, as some ciphertexts can be assigned negligible probabilities to equalize the ciphertext numbers across messages.
2. $\Omega_{\mathrm{EE}} = \Omega^{\mathrm{U,MD,KID}}_{\mathrm{PE}} \cup \Omega_{\mathrm{DE}}$. This follows from Proposition 9: every uniform, message-dependent, and key-independent probabilistic encryption scheme can be represented as an EE construction.
3. $\Omega^{\mathrm{KC}}_{\mathrm{E}} \subseteq \Omega^{\mathrm{M\text{-}IND}}_{\mathrm{E}}$. This means that encryption schemes achieving optimal security must be M-IND.
4. $I_{p_{\mathrm{m}}, p_{\mathrm{k}}}(\Omega_{\mathrm{HE}}) = I_{p_{\mathrm{m}}, p_{\mathrm{k}}}(\Omega^{\mathrm{U}}_{\mathrm{PE}}) = I_{p_{\mathrm{m}}, p_{\mathrm{k}}}(\Omega_{\mathrm{E}}) = 0$. This follows trivially, as a HE scheme with an MS-IND encoder achieves optimal security.

Refined from the insights in Section 5, we have:

1. For general non-uniform message and key distributions, it typically holds that $\mathbf{Adv}^{\mathrm{mrs}}_{\Omega_{\mathrm{EE}},p_{\mathrm{m}},p_{\mathrm{k}}} > p_{\max}$. Moreover, it generally holds that $\Omega_{\mathrm{E}}^{\mathrm{MC}} \cap \Omega_{\mathrm{PE}}^{\mathrm{U}} \subseteq \Omega_{\mathrm{PE}}^{\mathrm{U,KD}}$. This implies that uniform probabilistic encryption with optimal message confidentiality must usually be key-dependent. Similar to KC, without the constraint of uniformity, this proposition does not hold.
2. $\mathbf{Adv}^{\mathrm{mrs}}_{\Omega_{\mathrm{PE}}^{\mathrm{U}},p_{\mathrm{m}},p_{\mathrm{k}}} = p_{\max}$, as our $\mathsf{OE} \in \Omega_{\mathrm{PE}}^{\mathrm{U}}$ achieves the optimal security.